Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

`thin(Q) ≜ ⊇ ∩ (∈ \ (Q° ∈))` — `thin(Q)` takes a set `ys` to a subset `xs` that keeps a `Q`-lower bound of every `a∈ys`.

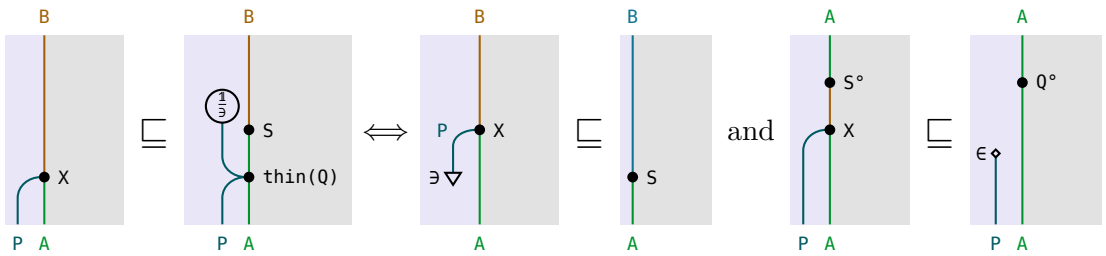
`thin(≤) ys thin(≤) xs⇔xs⊆ys∧(∀a∈ys. ∃b∈xs. b≤a).`

(8.1a)

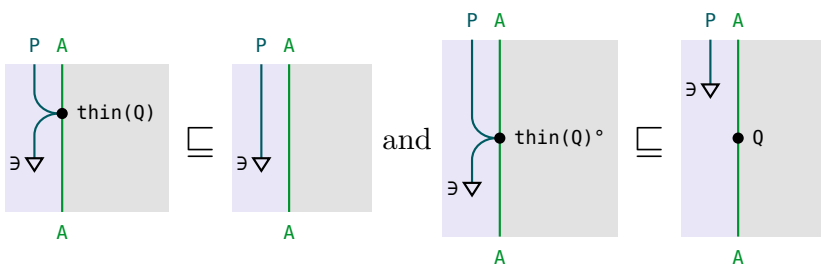
	the law	what it says
1	`X ⊑ S⫽∋ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈`	`S` is the whole algebra: it produces every candidate that `thin(Q)` thins. Everything kept is an `S`-value, and every `S`-value has a `Q`-lower bound among the kept ones
2	`thin(Q) ∋ ⊑ ∋`	everything a thinning keeps was in the set
3	`thin(Q)° ∋ ⊑ ∋ Q`	every element of the set has a `Q`-lower bound among the kept ones
4	`Q ⊑ R ⇒ thin(Q) ⊑ thin(R)`	the fewer pairs `Q` relates, the fewer subsets count as thinnings
5	`𝟙 ⊑ Q ⇒ 𝟙 ⊑ thin(Q)` `Q² ⊑ Q ⇒ thin(Q)² ⊑ thin(Q)`	keeping everything is always a legal thinning
6	`Q ⊑ R ∧ 𝟙 ⊑ Q ∧ R² ⊑ R ⇒ est(R) = thin(Q) est(R) 𝟙 ⊑ Q` `⇒ est(R) ⊑ thin(Q) est(R)`	**thin-introduction**: thinning first cannot lose an `R`-minimum
7	`thin(Q)⊒est(Q) 𝟙⫽∋` (8.2) — (8.2) `est(Q) 𝟙⫽∋ ⊑ thin(Q)`	**thin-elimination**: keeping one element is a thinning, but its domain is the sets `est(Q)` is defined on
8	`S⫽∋ thin(Q)⊒ S⫽∋ est(R) 𝟙⫽∋` (8.3), `R∩(S°S)⊑Q` — thin-elimination variant	the usable variant: `R` need only refine `Q` between values `S` gives one argument
9	`P(thin(Q)) union ⊑ union thin(Q)` (8.4) — (8.4) `P(thin(Q)) union ⊑ union thin(Q)`	thinning each member set is a thinning of the union

§8.1.1. `X ⊑ S⫽∋ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈` — the definition of `S⫽∋ thin(Q)`

(8.1.1a)



(8.1.1b)

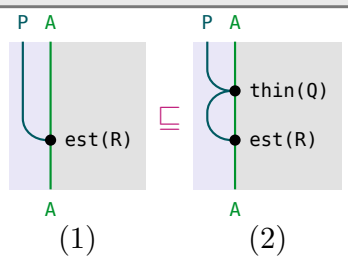


`X ⊑ S⫽∋ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈` at `X≜thin(Q)`, `S≜∋`: the left side is `thin(Q)⊑thin(Q)`, since `∋⫽∋=Λ(∋)=𝟙`

§8.1.2. $Q \sqsubseteq R \wedge \mathbb{1} \sqsubseteq Q \wedge R^2 \sqsubseteq R \Rightarrow \mathrm{est}(R) = \mathrm{thin}(Q)\ \mathrm{est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$\mathbb{1} \sqsubseteq Q \Rightarrow \mathrm{est}(R) \sqsubseteq \mathrm{thin}(Q)\ \mathrm{est}(R)$ (8.1.2a)

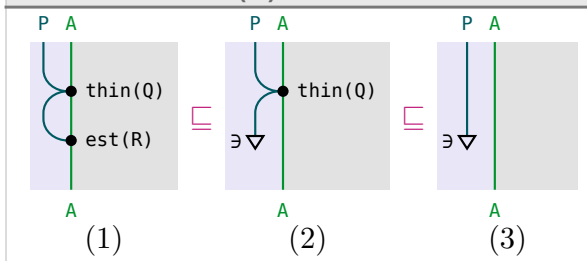
the $\sqsubseteq$ half: keeping everything is a thinning — $\mathbb{1} \sqsubseteq Q$



1	$\mathrm{est}(R)$	$\mathbb{1} \sqsubseteq \mathrm{thin}(Q)$ — $\mathbb{1} \sqsubseteq Q \Rightarrow \mathbb{1} \sqsubseteq \mathrm{thin}(Q)$, $Q^2 \sqsubseteq Q \Rightarrow \mathrm{thin}(Q)^2 \sqsubseteq \mathrm{thin}(Q)$
2	$\sqsubseteq \mathrm{thin}(Q)\ \mathrm{est}(R)$	

$\mathrm{thin}(Q)\ \mathrm{est}(R) \sqsubseteq \ni$ (8.1.2b)

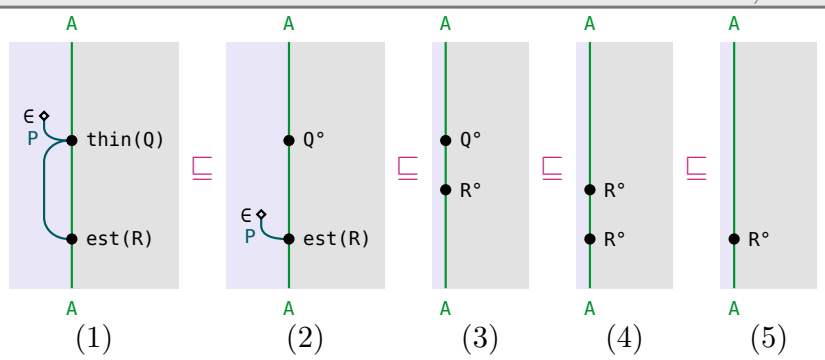
the $\sqsupseteq$ half, first condition of the UP of est at $X \triangleq \mathrm{thin}(Q)\ \mathrm{est}(R)$ — $X \sqsubseteq \mathrm{est}(R) \Leftrightarrow X \sqsubseteq \ni \wedge \in X \sqsubseteq R^\circ$



1	$\mathrm{thin}(Q)\ \mathrm{est}(R)$	$R \cap S \sqsubseteq R$
2	$\sqsubseteq \mathrm{thin}(Q) \ni$	$\mathrm{thin}(Q) \ni \sqsubseteq \ni$
3	$\sqsubseteq \ni$	

$Q \sqsubseteq R \wedge R^2 \sqsubseteq R \Rightarrow \in \mathrm{thin}(Q)\ \mathrm{est}(R) \sqsubseteq R^\circ$ (8.1.2c)

the $\sqsupseteq$ half, second condition



1	$\in \mathrm{thin}(Q)\ \mathrm{est}(R)$	$\in \mathrm{thin}(Q) \sqsubseteq Q^\circ\ \in$
2	$\sqsubseteq Q^\circ \in \mathrm{est}(R)$	$\in \mathrm{est}(R) \sqsubseteq R^\circ$
3	$\sqsubseteq Q^\circ\ R^\circ$	$Q \sqsubseteq R$
4	$\sqsubseteq R^{\circ 2}$	$R^2 \sqsubseteq R \Rightarrow R^{\circ 2} \sqsubseteq R^\circ$
5	$\sqsubseteq R^\circ$	

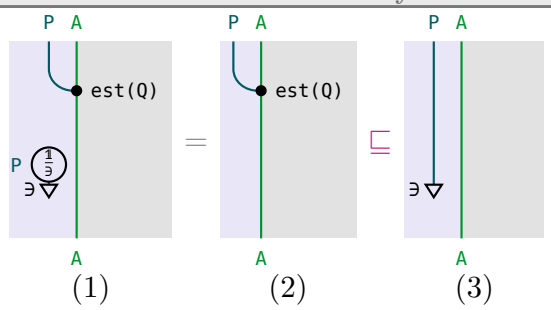
§8.1.3. Properties of thinning

est(Q) $\frac{1}{\ni}$ ⊑ thin(Q) (8.2)

the singleton holding a Q-least member of a set is a thinning of that set

est(Q) $\frac{1}{\ni}$ ∋ ⊑ ∋

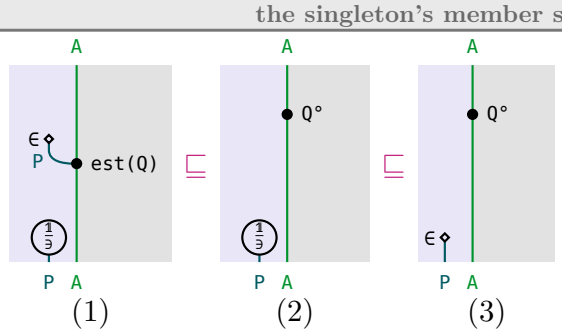
every member of the singleton is a member of the set — ≜ **thin**, ∋/∋ half



1	est(Q) $\frac{1}{\ni}$ ∋	$\frac{1}{\ni}$ ∋ = 𝟙
2	= est(Q)	R ∩ S ⊑ R
3	⊑ ∋	

∈ est(Q) $\frac{1}{\ni}$ ⊑ Q° ∈

the singleton's member stands in Q to every member of the set — ≜ **thin**, ∈\(Q°∈) half

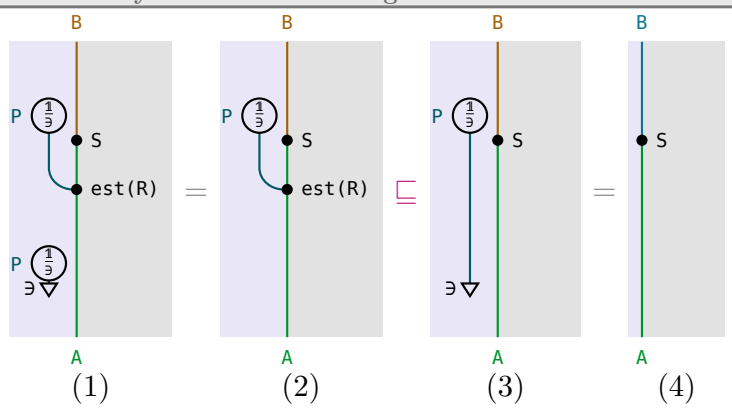


1	∈ est(Q) $\frac{1}{\ni}$	∈ est(R) ⊑ R°
2	⊑ Q° $\frac{1}{\ni}$	$\frac{1}{\ni}$ ⊑ ∈
3	⊑ Q° ∈	

$$\mathtt{R} \cap (\mathtt{S}^\circ\ \mathtt{S}) \sqsubseteq \mathtt{Q} \Rightarrow \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}\ \tfrac{\mathbb{1}}{\ni} \sqsubseteq \tfrac{\mathtt{S}}{\ni}\ \mathtt{thin(Q)} \tag{8.3}$$

$$\tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}\ \tfrac{\mathbb{1}}{\ni} \ni \sqsubseteq \mathtt{S}$$

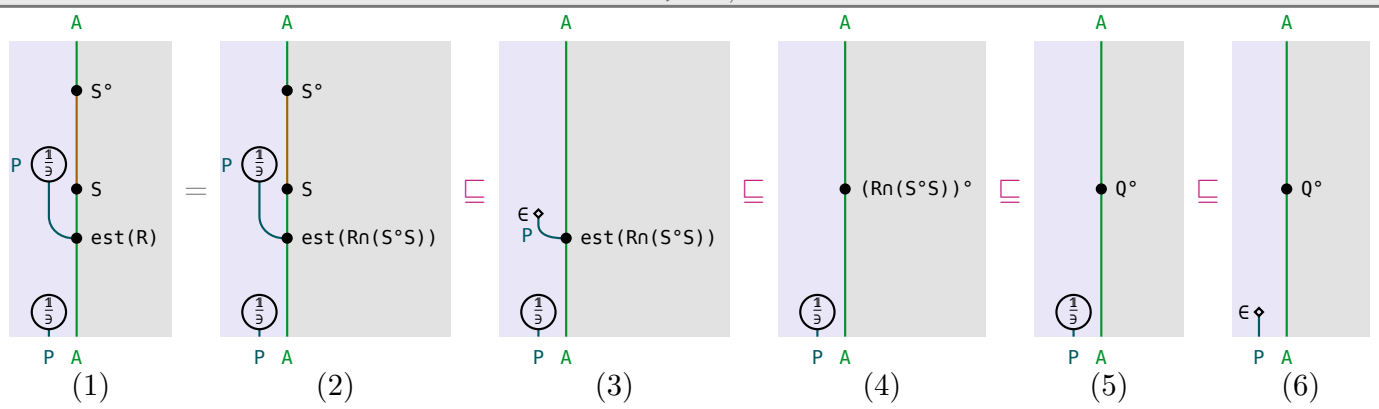
every member of the singleton is a value S returns — $\mathtt{X} \sqsubseteq \tfrac{\mathtt{S}}{\ni}\ \mathtt{thin(Q)} \Leftrightarrow \mathtt{X} \ni \sqsubseteq \mathtt{S} \wedge \mathtt{S}^\circ\ \mathtt{X} \sqsubseteq \mathtt{Q}^\circ \in$, first condition



1	$\tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}\ \tfrac{\mathbb{1}}{\ni} \ni$	$\tfrac{\mathbb{1}}{\ni} \ni = \mathbb{1}$
2	$= \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}$	$\mathtt{R} \cap \mathtt{S} \sqsubseteq \mathtt{R}$
3	$\sqsubseteq \tfrac{\mathtt{S}}{\ni} \ni$	$\tfrac{\mathtt{R}}{\ni} \ni = \mathtt{R}$
4	$= \mathtt{S}$	

$$\mathtt{R} \cap (\mathtt{S}^\circ\ \mathtt{S}) \sqsubseteq \mathtt{Q} \Rightarrow \mathtt{S}^\circ\ \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}\ \tfrac{\mathbb{1}}{\ni} \sqsubseteq \mathtt{Q}^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $\mathtt{X} \sqsubseteq \tfrac{\mathtt{S}}{\ni}\ \mathtt{thin(Q)} \Leftrightarrow \mathtt{X} \ni \sqsubseteq \mathtt{S} \wedge \mathtt{S}^\circ\ \mathtt{X} \sqsubseteq \mathtt{Q}^\circ \in$, second condition



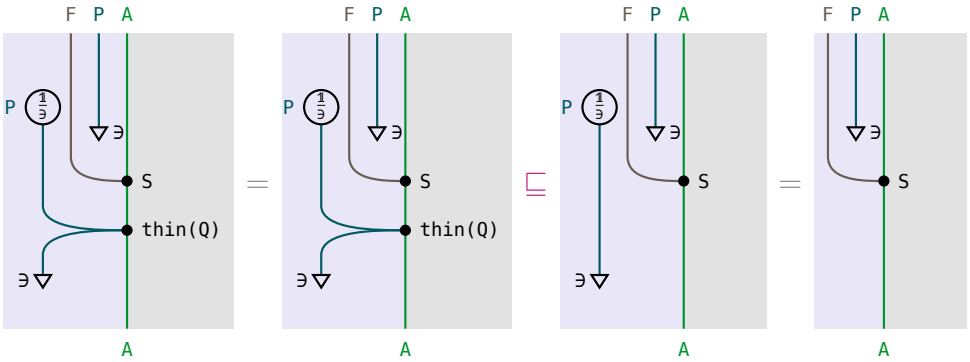
1	$\mathtt{S}^\circ\ \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}\ \tfrac{\mathbb{1}}{\ni}$	$\tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R} \cap (\mathtt{S}^\circ\ \mathtt{S})) = \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R)}$
2	$= \mathtt{S}^\circ\ \tfrac{\mathtt{S}}{\ni}\ \mathtt{est(R} \cap (\mathtt{S}^\circ\ \mathtt{S}))\ \tfrac{\mathbb{1}}{\ni}$	$\mathtt{T}^\circ\ \tfrac{\mathtt{T}}{\ni} \sqsubseteq \in$
3	$\sqsubseteq \in \mathtt{est(R} \cap (\mathtt{S}^\circ\ \mathtt{S}))\ \tfrac{\mathbb{1}}{\ni}$	$\in \mathtt{est(R)} \sqsubseteq \mathtt{R}^\circ$
4	$\sqsubseteq (\mathtt{R} \cap (\mathtt{S}^\circ\ \mathtt{S}))^\circ\ \tfrac{\mathbb{1}}{\ni}$	$\mathtt{R} \cap (\mathtt{S}^\circ\ \mathtt{S}) \sqsubseteq \mathtt{Q}$
5	$\sqsubseteq \mathtt{Q}^\circ\ \tfrac{\mathbb{1}}{\ni}$	$\tfrac{\mathbb{1}}{\ni} \sqsubseteq \in$
6	$\sqsubseteq \mathtt{Q}^\circ \in$	

preorder(Q) ∧ S Pres Q ⇒ (|$\frac{F(∋)\ S}{∋}$ thin(Q)|) ⊑ $\frac{(|S|)}{∋}$ thin(Q) Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

(|$\frac{F(∋)\ S}{∋}$ thin(Q)|) ∋ ⊑ (|S|)

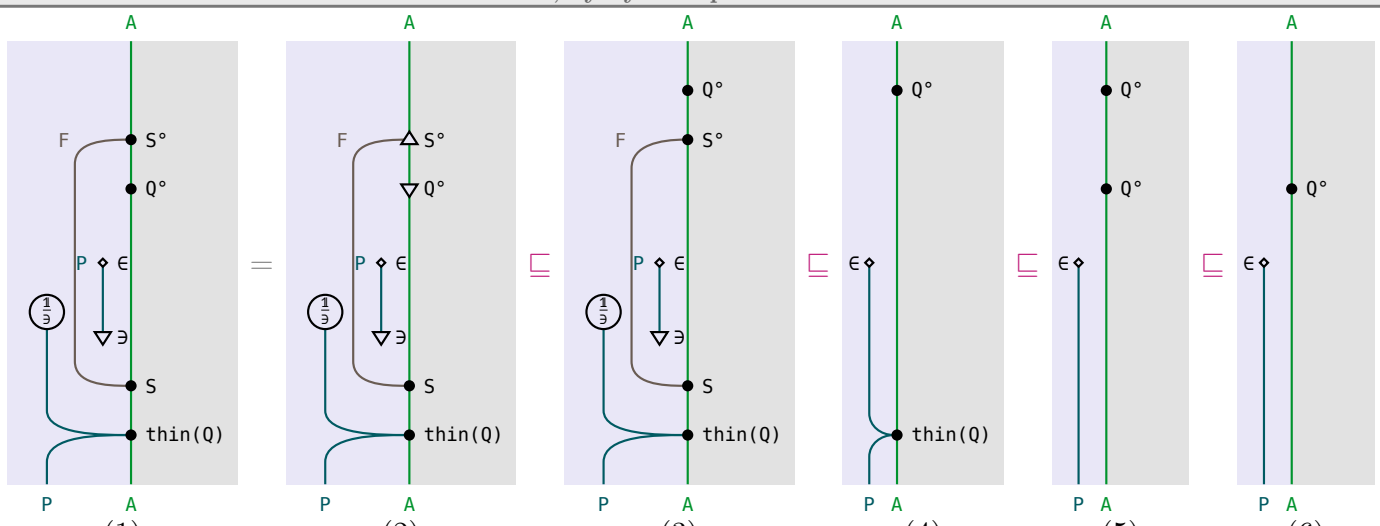
everything the thinning fold keeps is a value of the plain fold — **X ⊑ $\frac{S}{∋}$ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈** , first condition, by fusion from the chain below



1	$\frac{F(∋)\ S}{∋}$ thin(Q) ∋	
2	= $\frac{F(∋)\ S}{∋}$ thin(Q) ∋	thin(Q) ∋ ⊑ ∋
3	⊑ $\frac{F(∋)\ S}{∋}$ ∋	$\frac{R}{∋}$ ∋ = R
4	= F(∋) S	

preorder(Q) ∧ S Pres Q ⇒ (|S|)° (|$\frac{F(∋)\ S}{∋}$ thin(Q)|) ⊑ Q° ∈

every value of the plain fold is Q-above one the thinning fold keeps — **X ⊑ $\frac{S}{∋}$ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈** , second condition, by hylomorphism theorem from the chain below



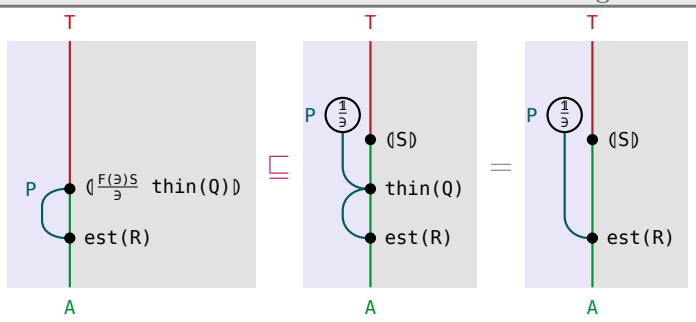
1	S° F(Q° ∈) $\frac{F(∋)\ S}{∋}$ thin(Q)	
2	= S° F(Q°) F(∈) $\frac{F(∋)\ S}{∋}$ thin(Q)	S Pres Q ⇒ S° F(Q°) ⊑ Q° S°
3	⊑ Q° S° F(∈) $\frac{F(∋)\ S}{∋}$ thin(Q)	S° F(∈) $\frac{F(∋)\ S}{∋}$ ⊑ ∈
4	⊑ Q° ∈ thin(Q)	∈ thin(Q) ⊑ Q° ∈
5	⊑ Q°² ∈	
6	⊑ Q° ∈	

(8.1.3a)

	step	type	example	what it does
1	S:F(A)→A is 8.1's algebra; example F(X)=V+V×X, S=[τ,cons], candidate a=cons(v,p)			
2	S°	A→FA	cons(v,p) ↦ r(v,p)	takes a apart into the input of its last step
3	F(Q°∈)	FA→F(PA)	r(v,p) ↦ r(v,ps), some p'∈ps with p' Q p	replaces each recursive component by a set holding a Q-no-worse one; v stays
4	$\frac{F(∋)S}{∋}$	F(PA)→PA	r(v,ps) ↦ {cons(v,t) | t∈ps}	picks one element from each set, combines by S, collects every result
5	thin(Q)	PA→PA	{cons(v,t) | t∈ps} ↦ ys	keeps a subset; every dropped element is Q-beaten by a kept one
6	⊑ Q°∈	A→PA	ys holds y with y Q cons(v,p') Q cons(v,p) = a	the thinned set still holds an element Q-no-worse than a (S monotonic on Q, Q transitive)

Q ⊑ R ∧ preorder(Q) ∧ preorder(R) ∧ S Pres Q ⇒ (⦇F(∋) S / ∋ thin(Q)⦈ est(R) ⊑ ⦇S⦈ / ∋ est(R) Corollary 8.1

the thinning fold refines the optimisation problem itself



(1) (2) (3)

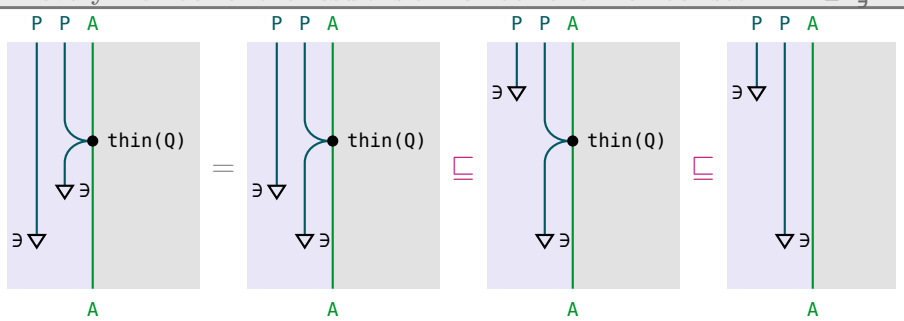
1	⦇F(∋) S / ∋ thin(Q)⦈ est(R)	Theorem 8.1 preorder(Q) ∧ S Pres Q ⇒ ⦇F(∋) S / ∋ thin(Q)⦈ ⊑ ⦇S⦈ / ∋ thin(Q)
2	⊑ ⦇S⦈ / ∋ thin(Q) est(R)	
3	= ⦇S⦈ / ∋ est(R)	

P(thin(Q)) union ⊑ union thin(Q) (8.4)

thinning each member set, then taking the union, is a thinning of the union

P(thin(Q)) union ∋ ⊑ ∋ ∋

every member of the result is a member of a member set — **X ⊑ S / ∋ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈** , first condition

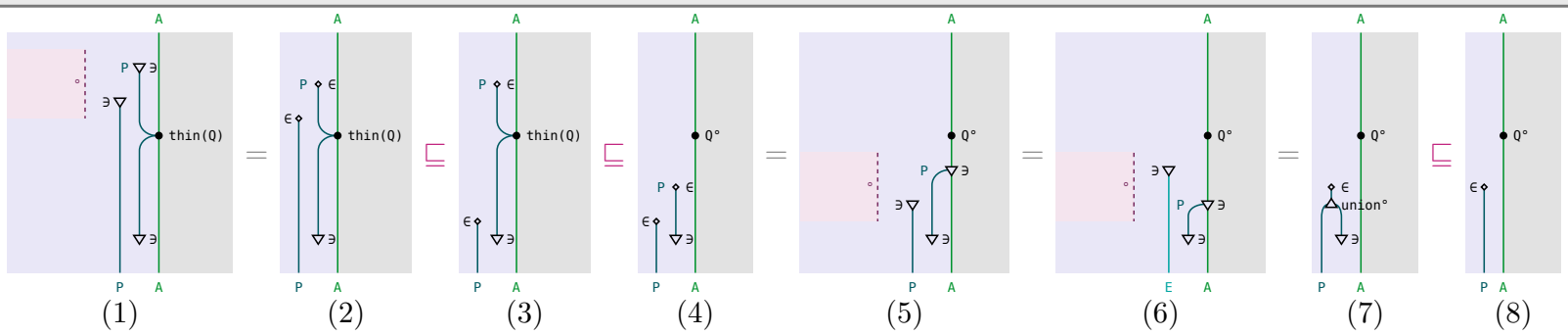


(1) (2) (3) (4)

1	P(thin(Q)) union ∋	union ∋ = ∋ ∋
2	= P(thin(Q)) ∋ ∋	P(R) ∋ ⊑ ∋ R
3	⊑ ∋ thin(Q) ∋	thin(Q) ∋ ⊑ ∋
4	⊑ ∋ ∋	

(∋ ∋)° P(thin(Q)) union ⊑ Q° ∈

every member of a member set has a **Q**-lower bound in the result — **X ⊑ S / ∋ thin(Q) ⇔ X ∋ ⊑ S ∧ S° X ⊑ Q° ∈** , second condition



(1) (2) (3) (4) (5) (6) (7) (8)

1	(∋ ∋)° P(thin(Q)) union	(R S)° = S° R°
2	= ∈ ∈ P(thin(Q)) union	∈ P(R) ⊑ R ∈
3	⊑ ∈ thin(Q) ∈ union	∈ thin(Q) ⊑ Q° ∈
4	⊑ Q° ∈ ∈ union	(R S)° = S° R°
5	= Q° (∋ ∋)° union	union ∋ = ∋ ∋
6	= Q° (union ∋)° union	(R S)° = S° R°
7	= Q° ∈ union° union	union = E(∋)
8	⊑ Q° ∈	

§8.2. Paths in a layered network

(8.2a)

	definition	type	meaning
1	`F(B,X) = B + (B × X)`	$\mathcal{A}\times\mathcal{A}\to\mathcal{A}$	a path is either one vertex a, or a vertex a followed by a path xs
2	`α = [τ,cons]`	`F(A,LA)→LA`	build a path: wrap one vertex as [a], or put a in front of xs
3	`wrapz ≜ ⟨τ,zero⟩`	`A→LA×ℕ`	a path of one vertex, no edge, cost 0
4	`(a,(xs,n)) ↦ ([a]⧺xs,wt(a,head(xs)) + n)`	`A×(LA×ℕ)→LA×ℕ`	put a in front of the path; the cost adds the weight of the new edge a→head(xs)
5	`cost ≜ ⦇[wrapz,consw]⦈ π₂`	`LA→ℕ`	the cost of $[a_0,\dots,a_n]$ is $wt(a_0,a_1)+\dots+wt(a_{n-1},a_n)$
6	`R ≜ cost ≤ cost°`	`LA→LA`	xs is cheaper than ys
7	`⦇F(∋,𝟙) α⦈ = L(∋)`	`LA→LB`	L(∋) relates a list of layers to every path that chooses one vertex from each layer
8	`minpath ⊑ L(∋)/∋ est(R)`	`L(PA)→LA`	the input $[x_0,\dots,x_n]$: L(PA) is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	`Q ≜ R ∩ (head head°)`	`LA→LA`	like R, and the two paths also start at the same vertex: head(xs) = head(ys)
10	`a ↦ l(a)`	`B→B+C`	mark a value as the left alternative
11	`b ↦ r(b)`	`C→B+C`	mark a value as the right alternative
12	`cpl ≜ cp(𝟙 × A)`	`PB×A→P(B×A)`	pair every element of the set with the value beside it
13	`cpr ≜ cp(A × 𝟙)`	`A×PB→P(A×B)`	pair the value beside the set with every element of the set
14	`F(∋,𝟙)/∋ = [P(l),cpl P(r)]`	`F(PB,X)→PF(B,X)`	cpl transpose: from layer x and a set of paths ps, choose a vertex a ∈ x
15	`F(𝟙,∋)/∋ = [𝟙/∋ P(l),cpr P(r)]`	`F(B,PX)→PF(B,X)`	cpr transpose: from a vertex a and a set of paths ps, choose a tail xs ∈ ps
16	`step ≜ cpr P(cons) est(R)`	`A×P(LA)→LA`	put a in front of every path in ps and keep a cheapest one

(8.2b)

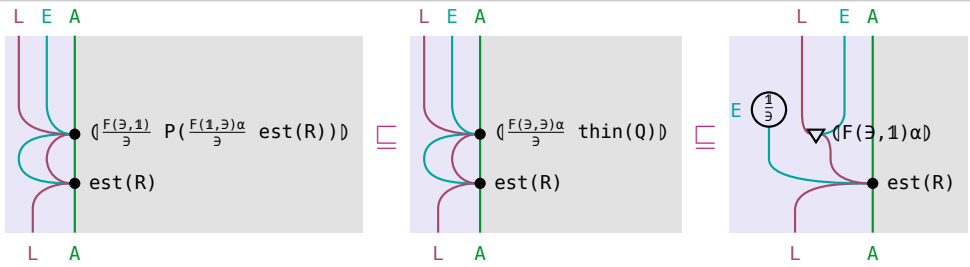
	the law	what it says
1	`F(∋,Q)α⊑F(∋,𝟙)αQ`	`F(∋,𝟙)α` is monotonic on `Q`; on `R` it is not, since the next edge can cost arbitrarily much
2	`S head⊑[𝟙,π₁]` `S≜F(𝟙,∋)α`	`S head` is simple, which gives `R∩(S°S)⊑Q`: between two paths `S` builds from one argument, equal cost and equal head already means `Q`

§8.2.1. Derivation

$$\left(\frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\ \mathtt{est(R)}\right)\right)\ \mathtt{est(R)} \sqsubseteq \frac{\left(F(\ni,1)\ \alpha\right)}{\ni}\ \mathtt{est(R)} \tag{8.2.1a}$$

a least-cost path in a layered network, as a fold over the layers

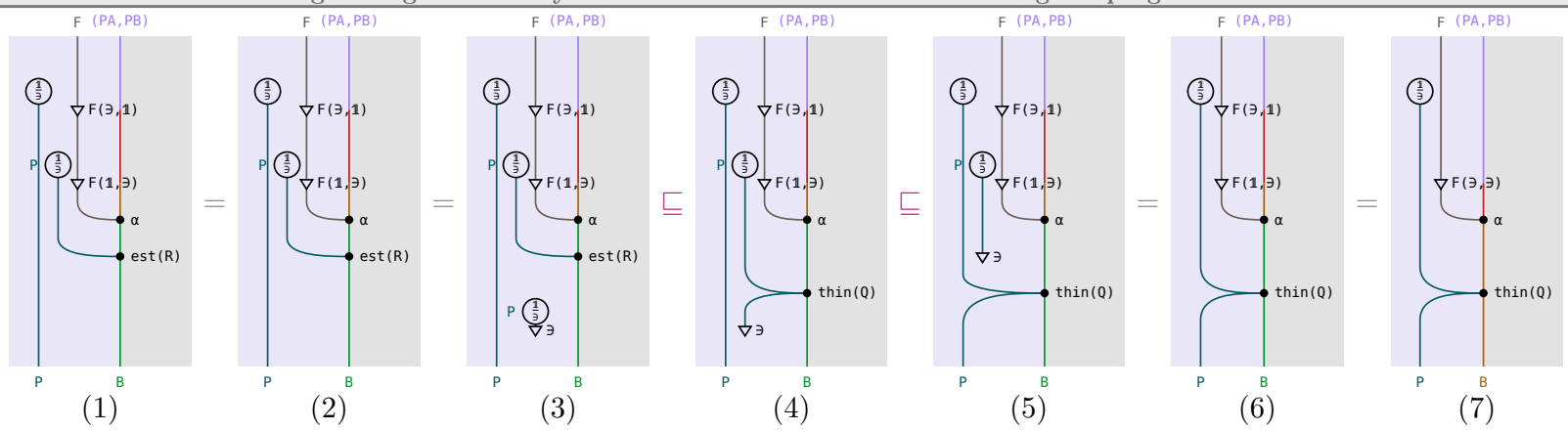
- Why it is needed: `est(R)` keeps one cheapest `w` of the set `Λ(S)(x)`; `thin(Q)` needs every dropped `z` to be `Q`-beaten by a kept one. For `{w}` alone to be a thinning we need `w Q z` for every `z` in `Λ(S)(x)`.
- We have `w R z` (`w` is cheapest) and `w (S°S) z` (both are built by `S` from the same argument `x`); the hypothesis turns the two into `w Q z`: among candidates from one argument, cheaper already means `Q`-better.



1	$\left(\frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right.\right.$ est(R))$\left.\right)$ est(R)	$\frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right.$ est(R)) $\sqsubseteq \frac{F(\ni,\ni)\ \alpha}{\ni}$ thin(Q)
2	$\sqsubseteq \left(\frac{F(\ni,\ni)\ \alpha}{\ni}\right.$ thin(Q)$\left.\right)$ est(R)	$\left(\frac{F(\ni,\ni)\ \alpha}{\ni}\right.$ thin(Q)$\left.\right)$ est(R) $\sqsubseteq \frac{\left(F(\ni,1)\ \alpha\right)}{\ni}$ est(R)
3	$\sqsubseteq \frac{\left(F(\ni,1)\ \alpha\right)}{\ni}$ est(R)	

$$\mathtt{S} \triangleq \mathtt{F(1,\ni)}\ \alpha\ ,\ \mathtt{R} \cap (\mathtt{S}^\circ\ \mathtt{S}) \sqsubseteq \mathtt{Q} \Rightarrow \frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\ \mathtt{est(R)}\right) \sqsubseteq \frac{F(\ni,\ni)\ \alpha}{\ni}\ \mathtt{thin(Q)} \tag{8.2.1b}$$

thinning the algebra of a layered network costs no more than taking the program's two cases



1	$\frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right.$ est(R))	
2	$= \frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right.$ est(R))	$P(\frac{1}{\ni})\ E(\ni) = \mathbb{1}$
3	$= \frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right.$ est(R) $\left.\frac{1}{\ni}\right)\ E(\ni)$	(8.3) R ∩ (S° S) ⊑ Q ⇒ $\frac{S}{\ni}$ est(R) $\frac{1}{\ni} \sqsubseteq \frac{S}{\ni}$ thin(Q)
4	$\sqsubseteq \frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right)\ P(\mathtt{thin(Q)})\ E(\ni)$	(8.4) P(thin(Q)) union ⊑ union thin(Q)
5	$\sqsubseteq \frac{F(\ni,1)}{\ni}\ P\left(\frac{F(1,\ni)\ \alpha}{\ni}\right)\ E(\ni)$ thin(Q)	$\frac{V\ S}{\ni} = \frac{V}{\ni}\ P(\frac{S}{\ni})$ union
6	$= \frac{F(\ni,1)\ F(1,\ni)\ \alpha}{\ni}$ thin(Q)	F(R,𝟙) F(𝟙,S) = F(R,S)
7	$= \frac{F(\ni,\ni)\ \alpha}{\ni}$ thin(Q)	

= cp(F) P(α) est(R) = [τ,step]

at the layered network (F(A,X)=A+A×X, B the paths V^+, α=[τ,cons]): $\frac{F(\mathbb{1},\ni)}{\ni}$ P(α) est(R)=[τ,step] — step ≜ cpr P(cons) est(R)

= $\frac{F(\ni,\mathbb{1})}{\ni}$ =𝟙+cpl — $\frac{F(\ni,\mathbb{1})}{\ni}$ = [P(l),cpl P(r)]

(8.2.1c)

```
F(A,X)   =  A   +   A  ×  X
            |       |     |
            R       R     S
            ↓       ↓     ↓
F(A',X') =  A'  +   A' ×  X'
```

`u F(R,S) v` holds iff one of:

- `u`, `v` are both `inl` and `d R d'`: the leaf moves by `R`;
- `u`, `v` are both `inr` and p_1 `R` q_1 ∧ p_2 `S` q_2: the vertex moves by `R`, the rest by `S`.
- An `inl` never relates to an `inr`: `R+S≜[R inl, S inr]` returns to the summand it came from, and `F(𝟙)=𝟙` allows no crossing.
- `F(∋,∋)` relates `inr({a,b},{p,q})` to `inr(a,p)`, `inr(a,q)`, `inr(b,p)`, `inr(b,q)`, and `inl({a,b})` to `inl(a)`, `inl(b)`.

§8.3. Implementing thin

(8.3a)

	definition	type	meaning
1	(setify ∋)(xs,x) ⇔ x ∈ xs	[A]→PA	the set of a list's elements
2	cup ≜ $\frac{(\pi_1\ \ni)\ \cup\ (\pi_2\ \ni)}{\ni}$	PA×PA→PA	the union of two sets
3	cp(F) ≜ $\frac{F(\ni)}{\ni}$	F(PA)→PFA	every F-shaped value whose parts are chosen from the given sets
4	cpLFn(zer,x) = [], cpLFn(PolyC.one,u) = [u, ()], cpLFn(const(a),c) = [c], cpLFn(arg₁,a) = [a], cpLFn(arg₂,xs) = xs , cpLFn(l ⊕ a,l(u)) = inl′(cpLFn(l,u)), cpLFn(a ⊕ r,r(v)) = inr′(cpLFn(r,v)), cpLFn(l ⊗ r,(u,v)) = cppFn(cpLFn(l,u),cpLFn(r,v))	relator(list(X))→ list(relator(X))	by the shape of F: a constant or arg₁ is a one-element list, arg₂ the list itself, a sum lists its summand, a product pairs every element of one list with every element of the other
5	linear(zer) = true, linear(one) = true, linear(const(a)) = true, linear(arg₁) = true, linear(arg₂) = true, linear(l ⊕ r) = linear(l) ∧ linear(r), linear(l ⊗ r) = linear(l) ∧ linear(r) ∧ (¬ hasArg₂(l) ∨ ¬hasArg₂(r))	P(PolyC(ι))	no product in F has the argument on both sides
6	sort(⩽) ≜ setify° ordered(⩽)	PA→LA	read a set back as one of its ⩽-ordered listings
7	(a,[]) ↦ [a] , bump(Q)((a,[b]⧺xs),ys) ⇔ Q(a,b) ∧ ys = [a]⧺xs ∨ Q(b,a) ∧ ys = [b]⧺xs ∨ ¬Q(a,b) ∧ ¬Q(b,a) ∧ ys = [a]⧺[b]⧺xs	A×[A]→[A]	put a in front of a thinned list, dropping whichever of a and the old head the other beats
8	thinL(Q) ≜ (setify thin(Q) setify°) ∩ subseq	[A]→[A]	only drop elements, and on the set of the elements be a thinning by Q
9	T ⊑ thinL(Q) ⇔ T ⊑ subseq ∧ T setify ⊑ setify thin(Q)		what a list implementation of thin(Q) must do: only drop elements, and thin the set
10	preorder(Q) ⇒ ◖[nil,bump(Q)]◗ ⊑ thinL(Q)		an implementation: thin a list in one pass, bumping each element into the thinned rest
11	minL(Q) ≜ setify est(Q)	[A]→A	a Q-least element of the list

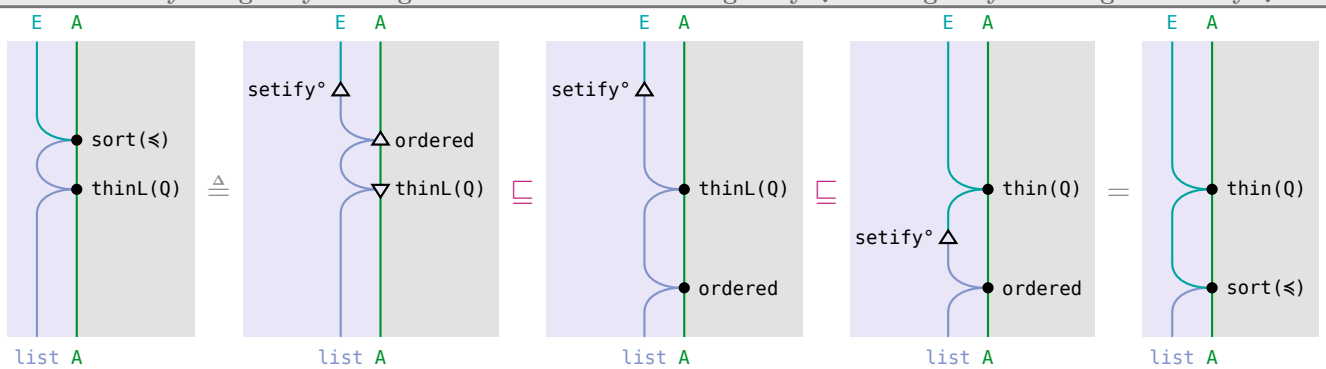
§8.3.1. Sorting sets

(8.3.1a)

	the law	what it says
1	`preorder(Q) ∧ connected(Q) ⇒ bumpFold(Q)([a]⧺xs,ys) ⇔ (∃w. minL(Q)([a]⧺xs,w) ∧ ys = [w])` (8.5)	what thinning should come to when it can: one element
2	`sort(⩽) thinL(Q) ⊑ thin(Q) sort(⩽)` (8.6) — (8.6) `sort(⩽) thinL(Q) ⊑ thin(Q) sort(⩽)`	the thinning may run after the sort
3	`sort(⩽) minL(R) ⊑ est(R)` (8.7)	a minimum of the sorted list is a minimum of the set
4	`sort(f ⩽ f°) f′ ⊑ P(f) sort(⩽)` (8.8)	shunt a function through a sort
5	`Coreflexive(p) ⇒ sort(⩽) filter(p) ⊑ E(p) sort(⩽)` (8.9) `p : A → A`	filtering a sorted list sorts the restricted set
6	`preorder(⩽) ∧ connected(⩽) ⇒ (sort(⩽) × sort(⩽)) merge(⩽) ⊑ cup sort(⩽)` (8.10)	merging two sorted lists sorts their union
7	`F(sort(⩽)) cpL ⊑ cp(F) sort(F(⩽))` (8.11), `F(A,X)=1+A×X`	`cpL` is the list implementation of the cartesian product `cp(F)`
8	`linear(F) ⇒ relator(sort(⩽)) cpL(F,A,X) ⊑ cp(relator(F,A)) sort(relator(⩽))` (8.11), `F` polynomial and linear	the same for every linear polynomial `F`

`sort(⩽) thinL(Q) ⊑ thin(Q) sort(⩽)` (8.6)

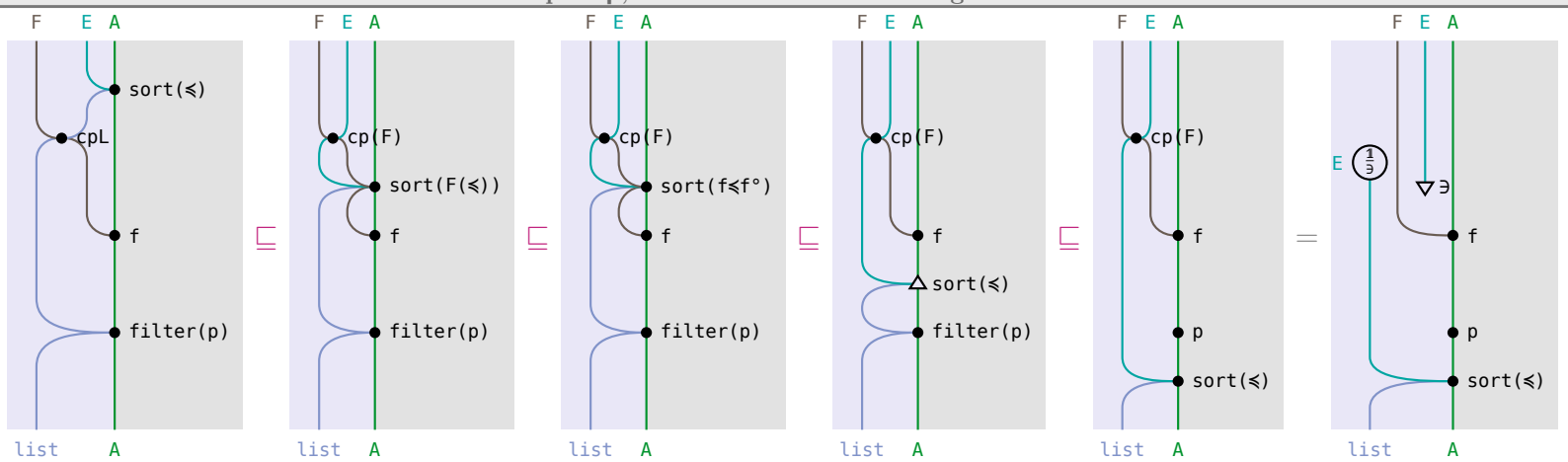
every list got by sorting the set and then thinning it by `Q` is also got by thinning the set by `Q` and then sorting.



1	`sort(⩽) thinL(Q)`	`sort(⩽) ≜ setify° ordered(⩽)`
2	`≜ setify° ordered thinL(Q)`	`ordered thinL(Q) ⊑ thinL(Q) ordered`
3	`⊑ setify° thinL(Q) ordered`	`setify° thinL(Q) ⊑ thin(Q) setify°`
4	`⊑ thin(Q) setify° ordered`	
5	`= thin(Q) sort(⩽)`	

Coreflexive(p) ∧ f Pres ⩽ ⇒ F(sort(⩽)) cpL f′ filter(p) ⊑ $\frac{\mathsf{F(∋)\ f\ p}}{\mathsf{∋}}$ sort(⩽) Lemma 8.1

sorting each argument set (**F(sort(⩽))**), forming every combination of their elements (**cpL**), applying **f** to each (**f′**) and keeping those that pass **p** (**filter(p)**) gives one of the sorted lists of the set of all **f(u)** that pass **p**, with **u** drawn from the argument sets.



1	`F(sort(⩽)) cpL f′ filter(p)`	`F(sort(⩽)) cpL ⊑ cp(F) sort(F(⩽))`
2	`⊑ cp(F) sort(F(⩽)) f′ filter(p)`	`Map(f)` `⇒ f Pres R ⇔ F(R) ⊑ f R f°`
3	`⊑ cp(F) sort(f ⩽ f°) f′ filter(p)`	`sort(f ⩽ f°) f′ ⊑ P(f) sort(⩽)`
4	`⊑ cp(F) P(f) sort(⩽) filter(p)`	`Coreflexive(p)` `⇒ sort(⩽) filter(p) ⊑ E(p) sort(⩽)`
5	`⊑ cp(F) P(f) E(p) sort(⩽)`	`Map(f)` `⇒ P(f) = E(f)`
6	`=` $\frac{\mathsf{F(∋)\ f\ p}}{\mathsf{∋}}$ `sort(⩽)`	

(8.3.1b)

At `F(A,X)=1+A×X`, `f(r(a,x))=a+x`, ⩽ is ≤, `p` passes the odd numbers, input `r(3,{4,1,2})`.

	left side	what it does	result
1	input	an `r(a,xs)`, `xs` a set	`r(3,{4,1,2})`
2	`F(sort(⩽))`	sorts the set in the `X` place; `a` is untouched	`r(3,[1,2,4])`
3	`cpL`	pairs `a` with each element, in list order	`[r(3,1),r(3,2),r(3,4)]`
4	`f′`	`f` on each	`[4,5,7]`
5	`filter(p)`	keeps the odd ones, order kept	`[5,7]`

	right side	what it does	result
1	input		`r(3,{4,1,2})`
2	`F(∋)`	relates `r(3,xs)` to each `r(3,x)` with `x∈xs`	`r(3,4)`, `r(3,1)`, `r(3,2)`
3	`f`		7, 4, 5
4	`p`	only odd values pass	7, 5
5	`Λ(…)`	collects every result into one set	`{5,7}`
6	`sort(⩽)`		`[5,7]`

- ⊑: the left's `[5,7]` is one result of the right; with ties `sort(⩽)` has several orders.
- `f` preserving ⩽ is used at `f′`: `[1,2,4]` is sorted, so `[4,5,7]` is; `filter(p)` only drops elements. With `f(r(a,x))=a−x` the left gives `[1,−1]` after `p`, not sorted, so ⊑ fails.

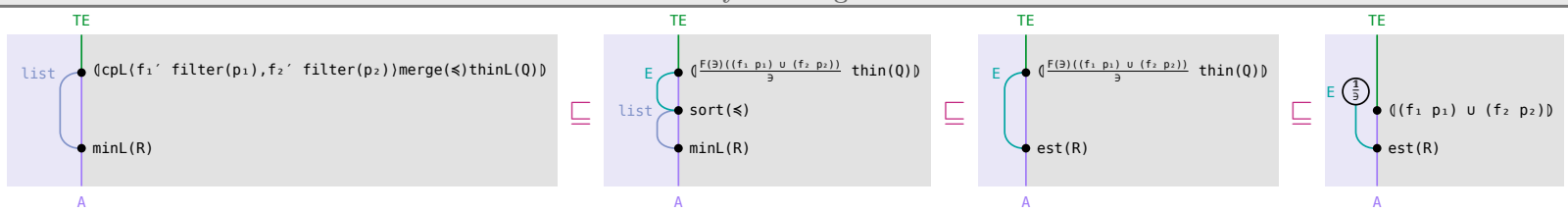
§8.3.2. Binary thinning

(8.3.2a)

Binary thinning data: `Q` a preorder with `Q⊑R` and both $f_1 p_1$, $f_2 p_2$ monotonic on `Q`; ⩽ a connected preorder with both f_1, f_2 monotonic on ⩽; `R` a preorder.

Coreflexive(p₁) ∧ Coreflexive(p₂) ∧ Q ⊑ R ∧ preorder(Q) ∧ preorder(R) ∧ (f₁ p₁) Pres Q ∧ (f₂ p₂) Pres Q ∧ preorder(⩽) ∧ connected(⩽) ∧ f₁ Pres ⩽ ∧ f₂ Pres ⩽ ⇒ ⦇cpL ⟨f₁′ filter(p₁),f₂′ filter(p₂)⟩ merge(⩽) thinL(Q)⦈ minL(R) ⊑ $\frac{⦇(f_1\ p_1)\ ∪\ (f_2\ p_2)⦈}{∋}$ est(R) Theorem 8.2

a fold on sorted lists of partial solutions, thinned at every step, refines the thinning specification — at binary thinning data.



(1) (2) (3) (4)

1	⦇cpL ⟨f₁′ filter(p₁),f₂′ filter(p₂)⟩ merge(⩽) thinL(Q)⦈ minL(R)	(8.3.2c) Coreflexive(p₁) ∧ Coreflexive(p₂) ∧ preorder(⩽) ∧ connected(⩽) ∧ f₁ Pres ⩽ ∧ f₂ Pres ⩽ ⇒ F(sort(⩽)) cpL ⟨f₁′ filter(p₁),f₂′ filter(p₂)⟩ merge(⩽) thinL(Q) ⊑ $\frac{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}{∋}$ thin(Q) sort(⩽)
2	⊑ ⦇$\frac{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}{∋}$ thin(Q)⦈ sort(⩽) minL(R)	sort(⩽) minL(R) ⊑ est(R)
3	⊑ ⦇$\frac{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}{∋}$ thin(Q)⦈ est(R)	S₁ Pres Q ∧ S₂ Pres Q ⇒ F(Q) (S₁ ∪ S₂) ⊑ (S₁ ∪ S₂) Q
4	⊑ $\frac{⦇(f_1\ p_1)\ ∪\ (f_2\ p_2)⦈}{∋}$ est(R)	

(8.3.2b)

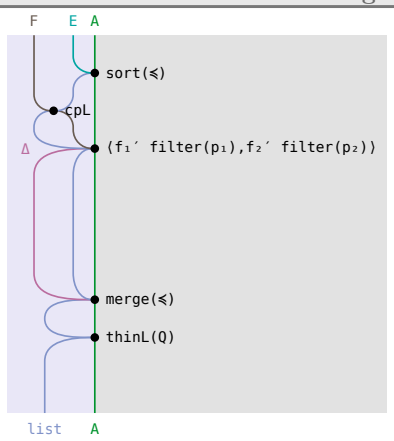
At F(A,X)=1+A×X, f₁(l())=f₂(l())=0, f₁(r(a,x))=x, f₂(r(a,x))=a+x, p₁ passes all, p₂ passes values ≤6, ⩽ is ≤, Q is =, R is ≥, input [3,3,1].

	input	f₁′ filter(p₁)	f₂′ filter(p₂)	merge(⩽)	thinL(Q)
1	l()	[0]	[0]	[0,0]	[0]
2	r(1,[0])	[0]	[1]	[0,1]	[0,1]
3	r(3,[0,1])	[0,1]	[3,4]	[0,1,3,4]	[0,1,3,4]
4	r(3, [0,1,3,4])	[0,1,3,4]	[3,4,6]	[0,1,3,3,4,4,6]	[0,1,3,4,6]
5	minL(R)				6

- merge(⩽) is one pass: both copies arrive ≤-sorted (fᵢ monotonic on ⩽, filter(pᵢ) only drops), so it only compares their two heads.
- thinL(Q) compares neighbours only; the list is ≤-sorted, so equal values are neighbours and one pass drops every repeat.

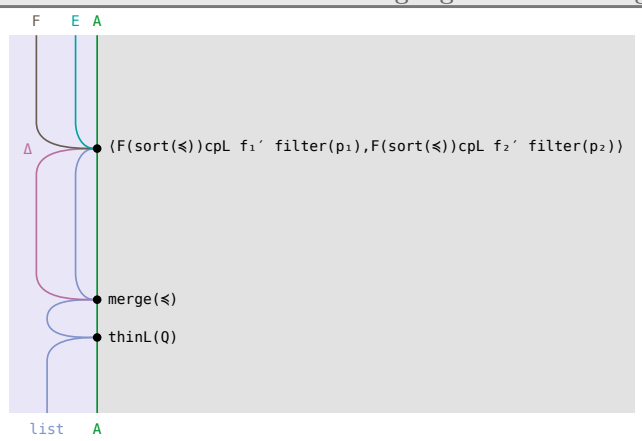
Coreflexive(p₁) ∧ Coreflexive(p₂) ∧ preorder(⩽) ∧ connected(⩽) ∧ f₁ Pres ⩽ ∧ f₂ Pres ⩽ ⇒
F(sort(⩽)) cpL ⟨f₁˙ filter(p₁),f₂˙ filter(p₂)⟩ merge(⩽) thinL(Q) ⊑ $\frac{\mathsf{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}}{∋}$
thin(Q) sort(⩽) (8.3.2c)

sorting the candidate set turns the thinning algebra into an algebra on sorted lists.



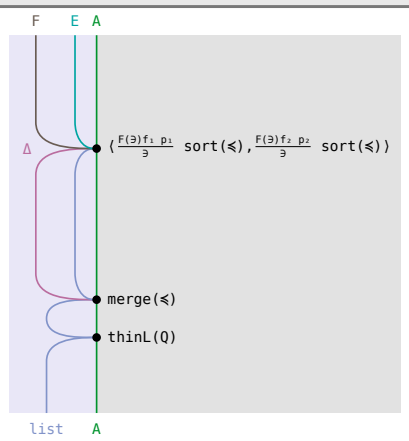
(1)

⊑



(2)

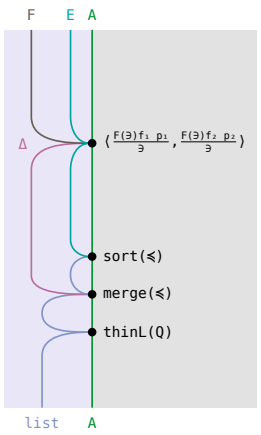
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(3)

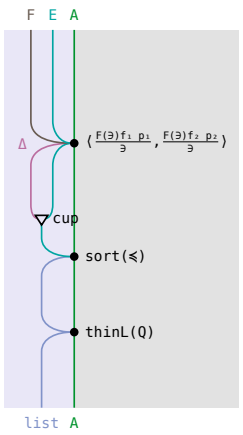
1	F(sort(⩽)) cpL ⟨f₁˙ filter(p₁),f₂˙ filter(p₂)⟩ merge(⩽) thinL(Q)	X ⟨R,S⟩ ⊑ ⟨X R,X S⟩
2	⊑ (F(sort(⩽)) cpL f₁˙ filter(p₁),F(sort(⩽)) cpL f₂˙ filter(p₂)) merge(⩽) thinL(Q)	Lemma 8.1 Coreflexive(p) ∧ f Pres ⩽ ⇒ F(sort(⩽)) cpL f˙ filter(p) ⊑ $\frac{\mathsf{F(∋)\ f\ p}}{∋}$ sort(⩽)
3	⊑ ⟨$\frac{\mathsf{F(∋)\ f_1\ p_1}}{∋}$ sort(⩽), $\frac{\mathsf{F(∋)\ f_2\ p_2}}{∋}$ sort(⩽)⟩ merge(⩽) thinL(Q)	⟨X,Y⟩ (R × S) = ⟨X R,Y S⟩

=



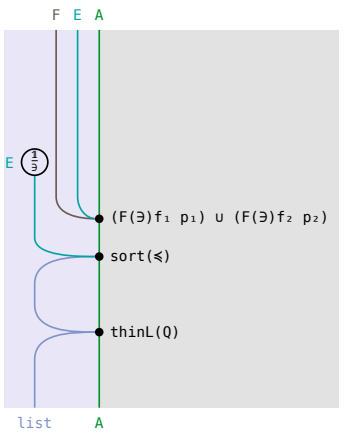
(4)

⊑



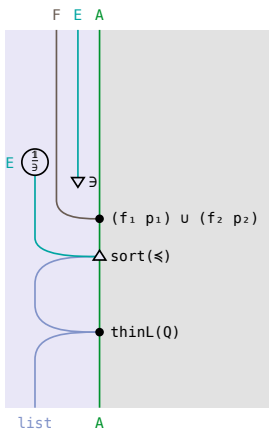
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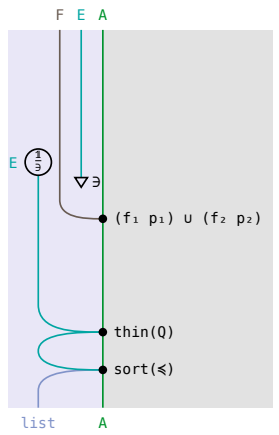
(6)

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(7)

⊑



(8)

4	= ⟨$\frac{\mathsf{F(∋)\ f_1\ p_1}}{∋}$, $\frac{\mathsf{F(∋)\ f_2\ p_2}}{∋}$⟩ (sort(⩽) × sort(⩽)) merge(⩽) thinL(Q)	preorder(⩽) ∧ connected(⩽) ⇒ (sort(⩽) × sort(⩽)) merge(⩽) ⊑ cup sort(⩽)
5	⊑ ⟨$\frac{\mathsf{F(∋)\ f_1\ p_1}}{∋}$, $\frac{\mathsf{F(∋)\ f_2\ p_2}}{∋}$⟩ cup sort(⩽) thinL(Q)	$\frac{\mathsf{R\ ∪\ S}}{∋}$ = ⟨$\frac{\mathsf{R}}{∋}$, $\frac{\mathsf{S}}{∋}$⟩ cup
6	= $\frac{\mathsf{(F(∋)\ f_1\ p_1)\ ∪\ (F(∋)\ f_2\ p_2)}}{∋}$ sort(⩽) thinL(Q)	R (S ∪ T) = (R S) ∪ (R T)
7	= $\frac{\mathsf{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}}{∋}$ sort(⩽) thinL(Q)	(8.6) sort(⩽) thinL(Q) ⊑ thin(Q) sort(⩽)
8	⊑ $\frac{\mathsf{F(∋)\ ((f_1\ p_1)\ ∪\ (f_2\ p_2))}}{∋}$ thin(Q) sort(⩽)	

§8.4. The knapsack problem

(8.4a)

	definition	type	meaning
1	FX = L + E × X	$\mathcal{A}\to\mathcal{A}$	here L ≜ 1 and E ≜ Item: the trivial base, or an item and what is built from the rest
2	total(f) ≜ f′ sum	Int	add up f over the items of the list
3	subseq ≜ ⦇[nil,cons ∪ π₂]⦈	[A]→[A]	every subsequence of the list
4	value ≜ val′ sum	[Item]→ Int	the total value of a packing
5	weight ≜ wt′ sum	[Item]→ Int	the total weight of a packing
6	[nil,cons] = [nil,cons]	F[Item]→ [Item]	keep the item: the list constructor
7	within(w,xs) ⇔ weight(xs) ≤ w	[Item]→ [Item]	the packing weighs at most w
8	S ≜ ([nil,cons] within(w)) ∪ [nil,π₂]	F[Item]→ [Item]	at each item, keep it if the packing still fits, or drop it
9	R ≜ value ≥ value°	[Item]→ [Item]	x is worth at least as much as y
10	Q(val,wt) ≜ R ∩ (weight ≤ weight°)	[Item]→ [Item]	x is worth at least as much as y and weighs no more

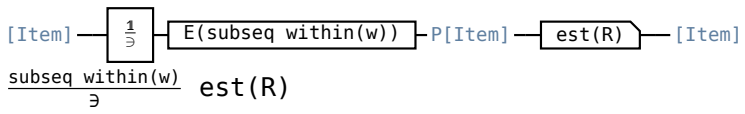
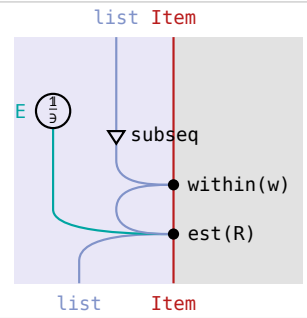
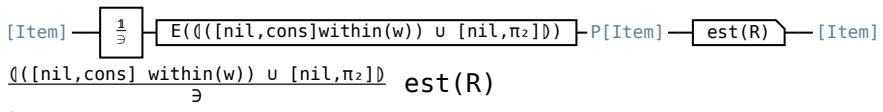
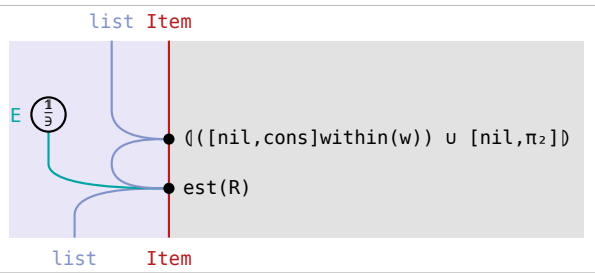
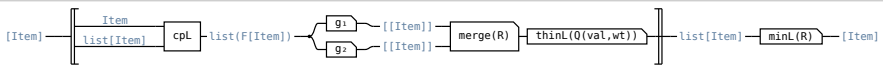
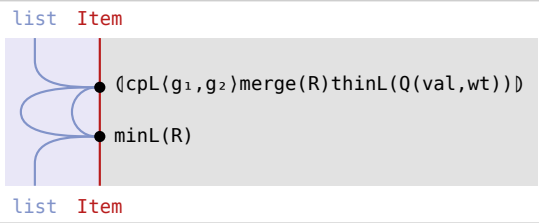
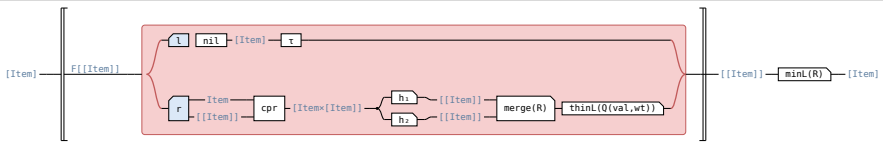
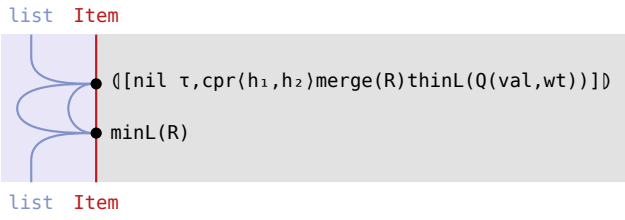
(8.4b)

	the law	what it says
1	(𝟙×R) (cons (within w))⊑cons (within w)R FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
2	F(Q(val,wt)) [nil,cons] within(w) ⊑ [nil,cons] within(w) Q(val,wt) F(Q(val,wt)) [nil,π₂] ⊑ [nil,π₂] Q(val,wt) ,	both halves are monotonic on Q once ties in value are broken by weight
3	F(R) [nil,cons] ⊑ [nil,cons] R F(R) [nil,π₂] ⊑ [nil,π₂] R	both algebras are monotonic on R itself, so R is the sort order P

§8.4.1. Derivation

0 ≤ w ∧ (∀i. 0 ≤ wt(i)) ⇒ ⦇cpL (g₁,g₂) merge(R) thinL(Q(val,wt))⦈ minL(R) ⊑ $\frac{\text{subseq within(w)}}{\ni}$ est(R) (8.4.1a)

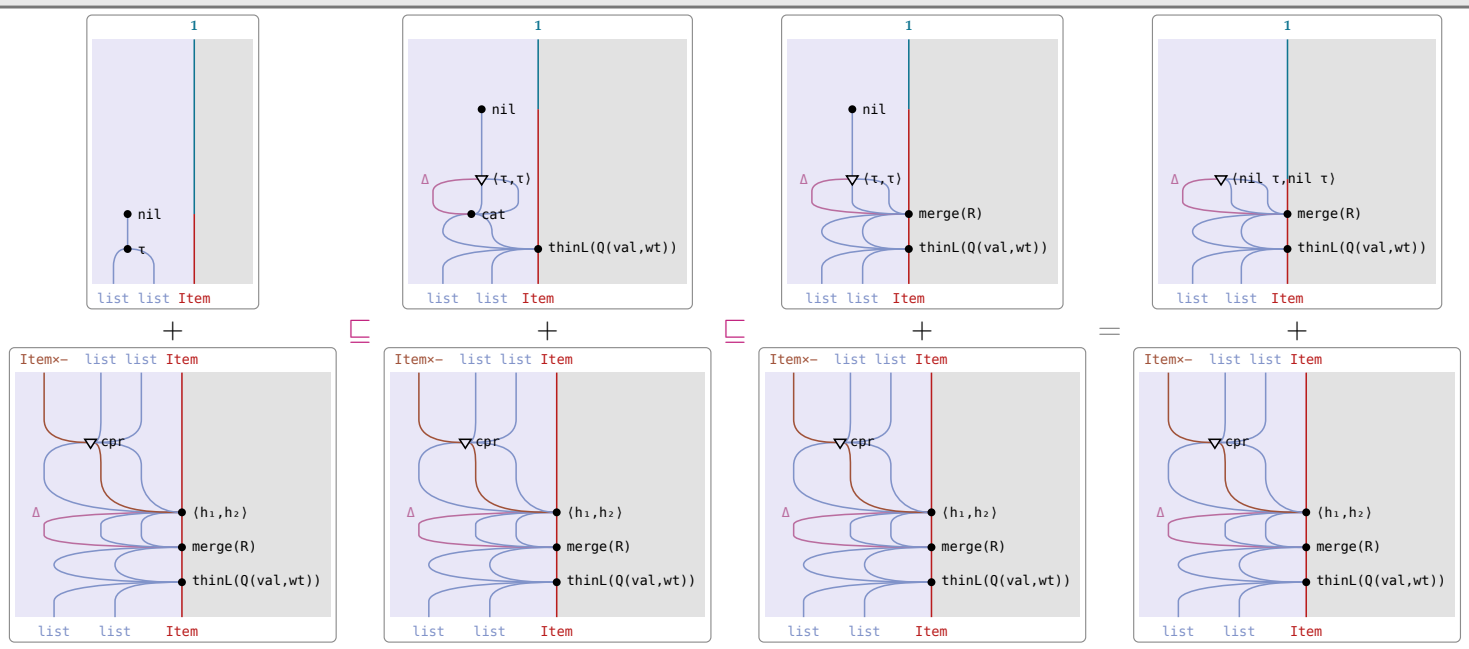
the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
 $\frac{\text{subseq within(w)}}{\ni}$ est(R)	
=  $\frac{\text{⦇[nil,cons] within(w)) ∪ [nil,π₂]⦈}}{\ni}$ est(R) fusion, weights non-negative.	
⊒  ⦇cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt))⦈ minL(R) binary thinning theorem, at P≜R, F linear, Q from F(Q(val,wt)) [nil,cons] within(w) ⊑ [nil,cons] within(w) Q(val,wt) , F(Q(val,wt)) [nil,π₂] ⊑ [nil,π₂] Q(val,wt)	
⊒  ⦇[nil τ,cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]⦈ minL(R) 0 ≤ w ⇒ [nil τ,cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))] ⊑ cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)) under the fold; minL(R) is head, packings coming out in descending value	

§8.4.2. Splitting the fold's body

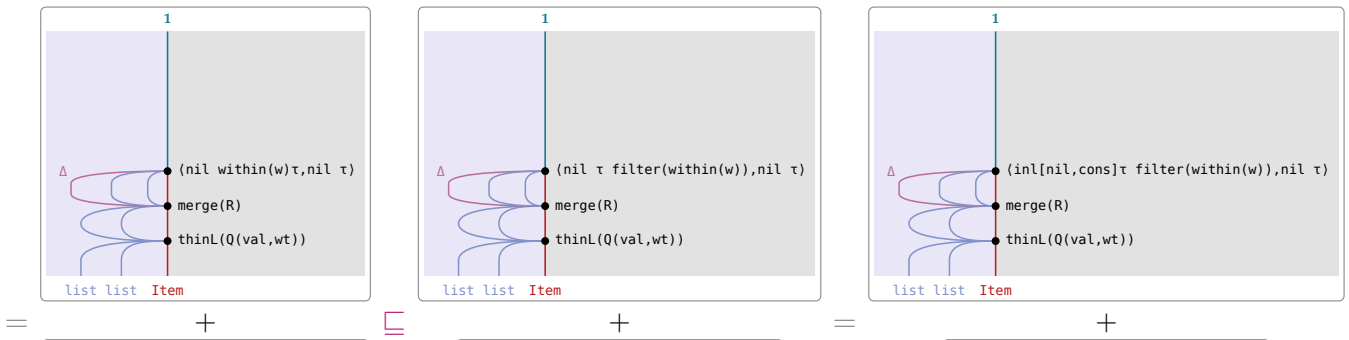
0 ≤ w ⇒ [nil τ,cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))] ⊑ cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)) (8.4.2a)

the fold's body splits into a base case **nil τ** and a step that extends each kept packing by the item or leaves it out



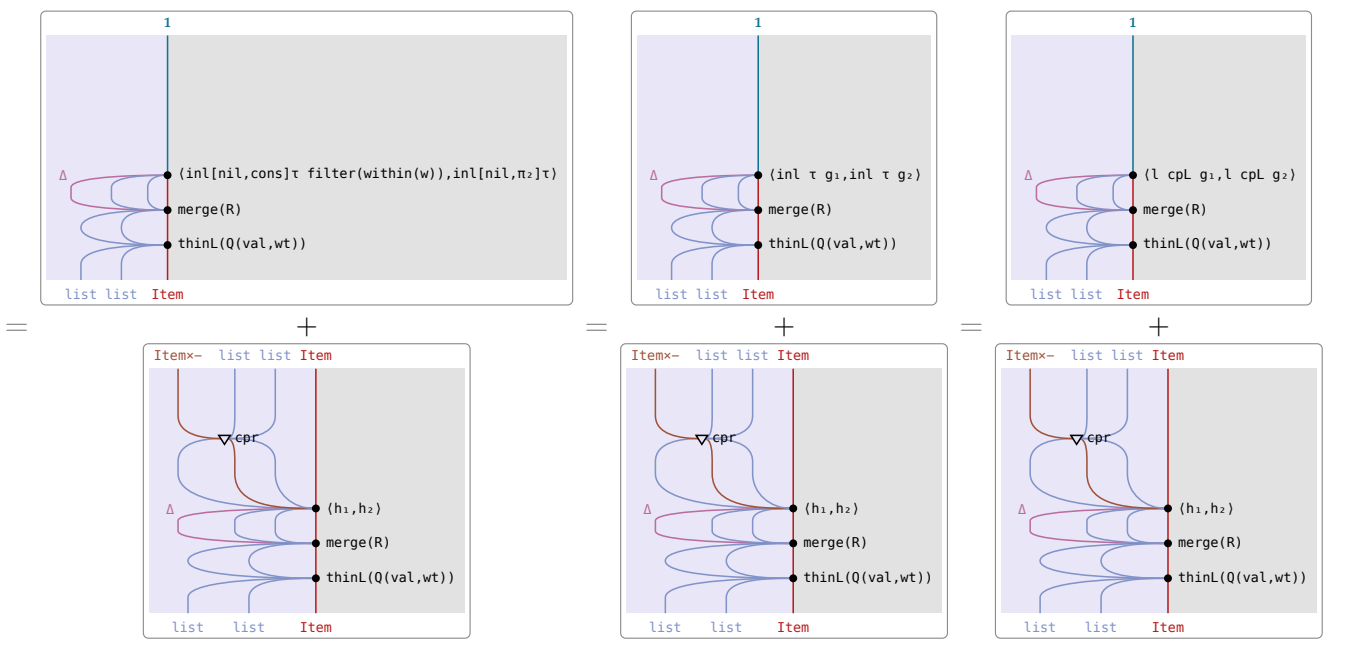
1	[nil τ,cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	1 ⊑ Q(val,wt)
2	⊑ [nil ⟨τ,τ⟩ cat thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	1 ⊑ R

3	⊑ [nil ⟨τ,τ⟩ merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩
4	= [(nil τ,nil τ) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	0 ≤ w ⇒ nil within(w) = nil



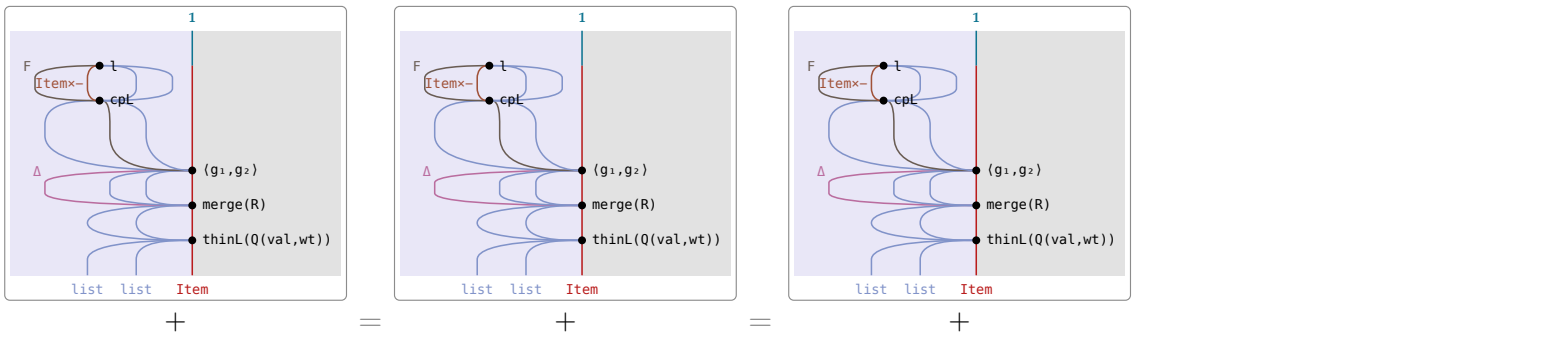
(5) (6) (7)

5	= [(nil within(w) τ,nil τ) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	Coreflexive(within(w))
6	⊑ [(nil τ filter(within(w)),nil τ) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	inl [nil,cons] = nil
7	= [(inl [nil,cons] τ filter(within(w)),nil τ) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	inl [nil,π₂] = nil

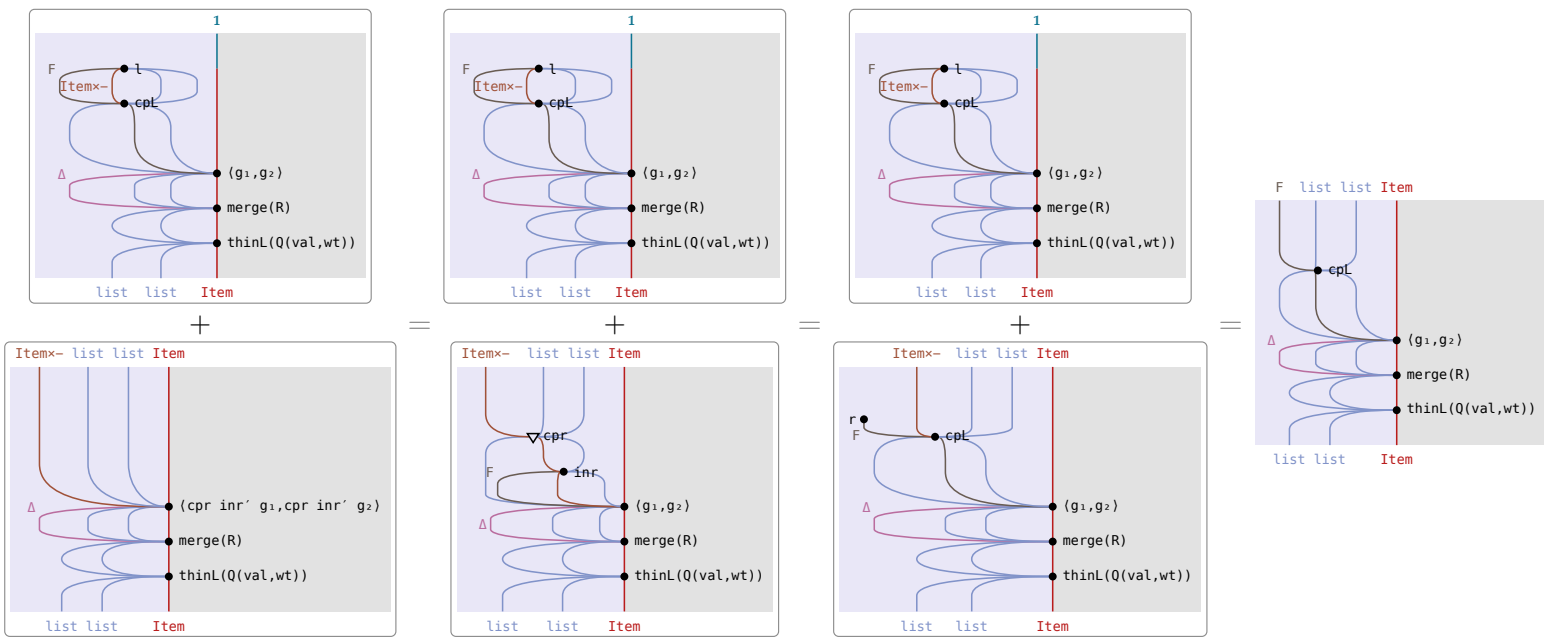


(8) (9) (10)

8	= [(inl [nil,cons] τ filter(within(w)),inl [nil,π₂] τ) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	τ R′ = R τ
9	= [(inl τ g₁,inl τ g₂) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	l cpL = inl τ
10	= [(l cpL g₁,l cpL g₂) merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩



11	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),cpr ⟨h₁,h₂⟩ merge(R) thinL(Q(val,wt))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩
12	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),⟨cpr h₁,cpr h₂⟩ merge(R) thinL(Q(val,wt))]	inr′ [nil,π₂]′ = π₂′
13	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),⟨cpr h₁,cpr inr′ g₂⟩ merge(R) thinL(Q(val,wt))]	inr′ [nil,cons]′ = cons′



14	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),⟨cpr inr′ g₁,cpr inr′ g₂⟩ merge(R) thinL(Q(val,wt))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩
15	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),cpr inr′ ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt))]	r cpL = cpr inr′
16	= [l cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt)),r cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt))]	l T = R ∧ r T = S ⇒ T = [R,S]
17	= cpL ⟨g₁,g₂⟩ merge(R) thinL(Q(val,wt))	

§8.5. The paragraph problem

(8.5a)

	definition	type	meaning
1	`FX = L + E × X`	`𝐴→𝐴`	here L ≜ Word and E ≜ Word: one word, or a word and what is built from the rest
2	`Line ≜ L(Word)`	`Type`	a line is a non-empty list of words
3	`Para ≜ L(Line)`	`Type`	a paragraph is a non-empty list of lines
4	`μF ≜ wrap(L) | cons(E,μF)`	`L→μF`	the one-element non-empty list holding a
5	`new(a,xss) ≜ [[a]]⧺xss`	`Para`	start a new first line holding only the word a
6	`glue(a,[xs]) = [[a]⧺xs], glue(a,[xs]⧺a') = [[a]⧺xs]⧺a'`	`Para→ Para`	put the word a at the front of the first line
7	`partition ≜ ⦇[τ²,new] ∪ [τ²,glue]⦈`	`L(Word)→ Para`	every way of breaking the words into lines
8	`width[a] = len(a), width([a]⧺xs) = (len(a) + width(xs)) + 1`	`Line→Int`	the lengths of the words plus one space between neighbours
9	`head[xs] = xs, head([xs]⧺a) = xs`	`Para→ Line`	the first line of the paragraph
10	`fits(w,[xs]) = width(xs) ≤ w, fits(w,[xs]⧺a) = width(xs) ≤ w ∧ fits(w,a)`	`Para→ Para`	every line is at most w wide
11	`ok(w,xss) ≜ width(head(xss)) ≤ w`	`Para→ Para`	the paragraphs whose first line is at most w wide
12	`sqr(n) ≜ nn`	`Int`	n squared
13	`waste(w,[xs]) = 0, waste(w,[xs]⧺a) = sqr(w−width(xs)) + waste(w,a)`	`Para→Int`	the squared white space left on every line but the last, added up
14	`R ≜ waste(w) ≤ waste(w)°`	`Para→ Para`	p wastes no more than q
15	`Q(len)(w) ≜ R ∩ (head head°)`	`Para→ Para`	p wastes no more than q and has the same first line
16	`start(a) ≜ [[[a]]]`	`[Para]`	the one candidate for a single word: the paragraph of one line holding that word

(8.5b)

	the law	what it says
1	`(𝟙×R) glue⊑glue R` FALSE	`glue` is not monotonic on `R`: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this

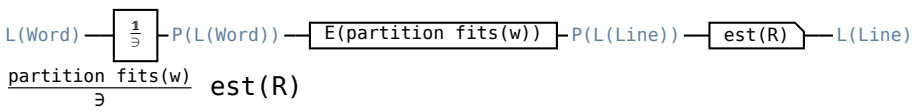
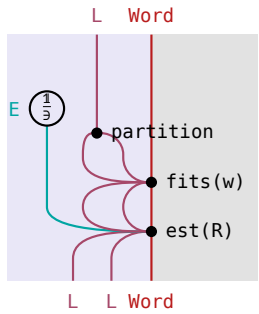
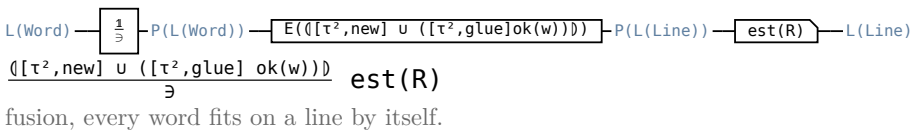
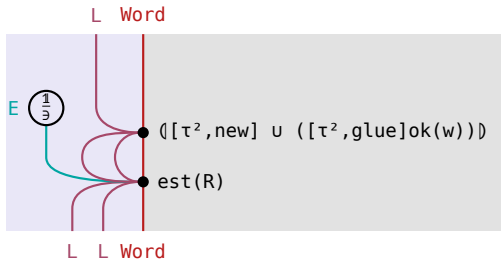
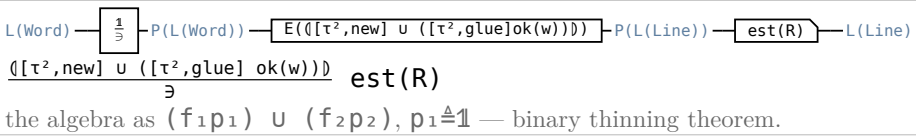
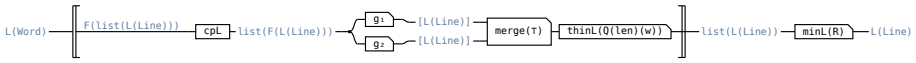
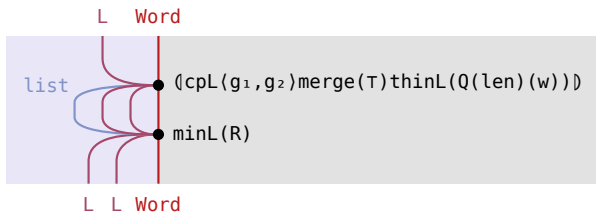
2	`F(Q(len)(w)) [τ²,new] ⊑ [τ²,new] Q(len)(w)`
	`(∀a. 0 ≤ len(a)) ⇒ F(Q(len)(w)) [τ²,glue] ok(w) ⊑ [τ²,glue] ok(w) Q(len)(w)`
	`cons` monotonic on `collect≤collect°.` ,
		both halves are monotonic on `Q` once ties in waste are broken by the first line

3	`merge(⊤) = cat`	`⊤` needs no sorting at all
4	`F(⊤) [τ²,new] ⊑ [τ²,new] ⊤` `F(⊤) [τ²,glue] ⊑ [τ²,glue] ⊤`	both algebras are monotonic on `⊤`, so `⊤` is the sort order `P`

§8.5.1. Derivation

(∀a. 0 ≤ len(a)) ∧ (∀a. len(a) ≤ w) ⇒ ⦇cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w))⦈ minL(R) ⊑ $\frac{\text{partition fits(w)}}{\ni}$ est(R) (8.5.1a)

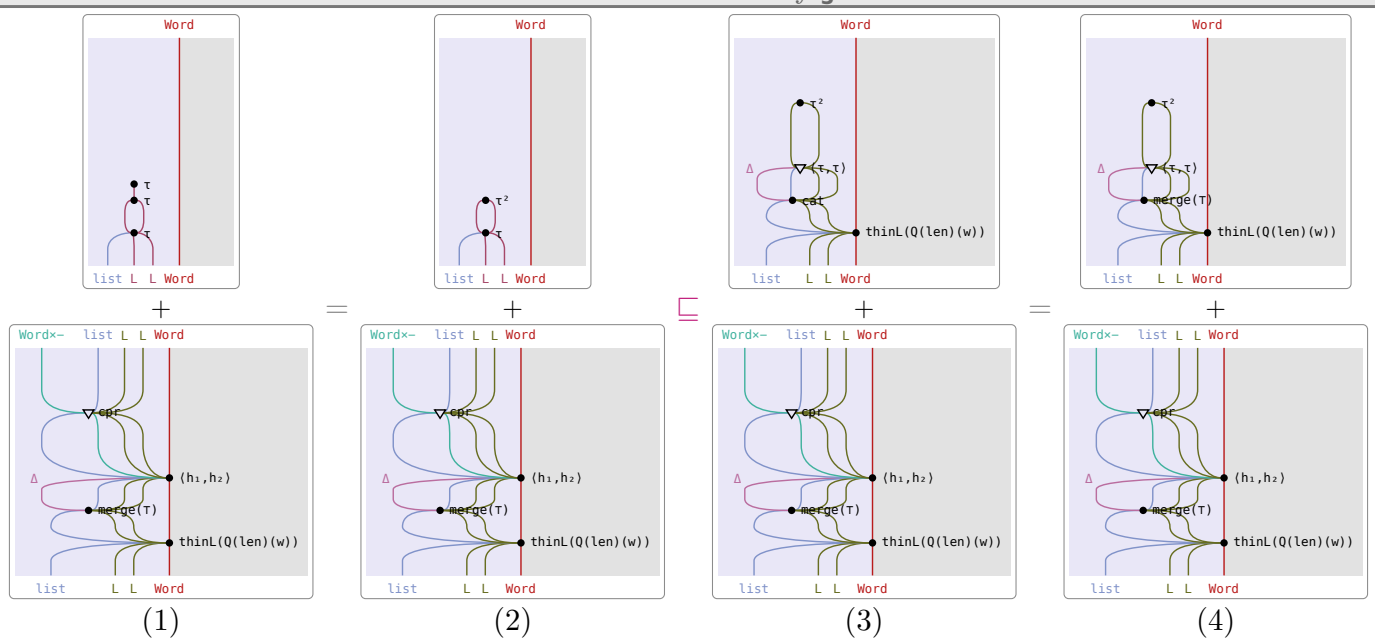
a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
 $\frac{\text{partition fits(w)}}{\ni}$ est(R)	
=  $\frac{\text{⦇[τ²,new] ∪ ([τ²,glue] ok(w))⦈}}{\ni}$ est(R) fusion, every word fits on a line by itself.	
=  $\frac{\text{⦇[τ²,new] ∪ ([τ²,glue] ok(w))⦈}}{\ni}$ est(R) the algebra as (f₁p₁) ∪ (f₂p₂), p₁≜𝟙 — binary thinning theorem.	
 ⦇cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w))⦈ minL(R) ⊒ binary thinning theorem, at P≜T with merge T=cat (merge(T) = cat), Q from F(Q(len)(w)) [τ²,new] ⊑ [τ²,new] Q(len)(w) , (∀a. 0 ≤ len(a)) ⇒ F(Q(len)(w)) [τ²,glue] ok(w) ⊑ [τ²,glue] ok(w) Q(len)(w)	
= ⦇[start,cpr ⟨h₁,h₂⟩ cat thinL(Q)]⦈ minL(R) cpL=τ+cpr, gᵢ along the coproduct — start(a) ≜ [[[a]]] , new(a,xss) ≜ [[a]]#xss , glue(a,[xs]) = [[a]#xs], glue(a,[xs]#a') = [[a]#xs]#a'	

§8.5.2. Splitting the fold's body

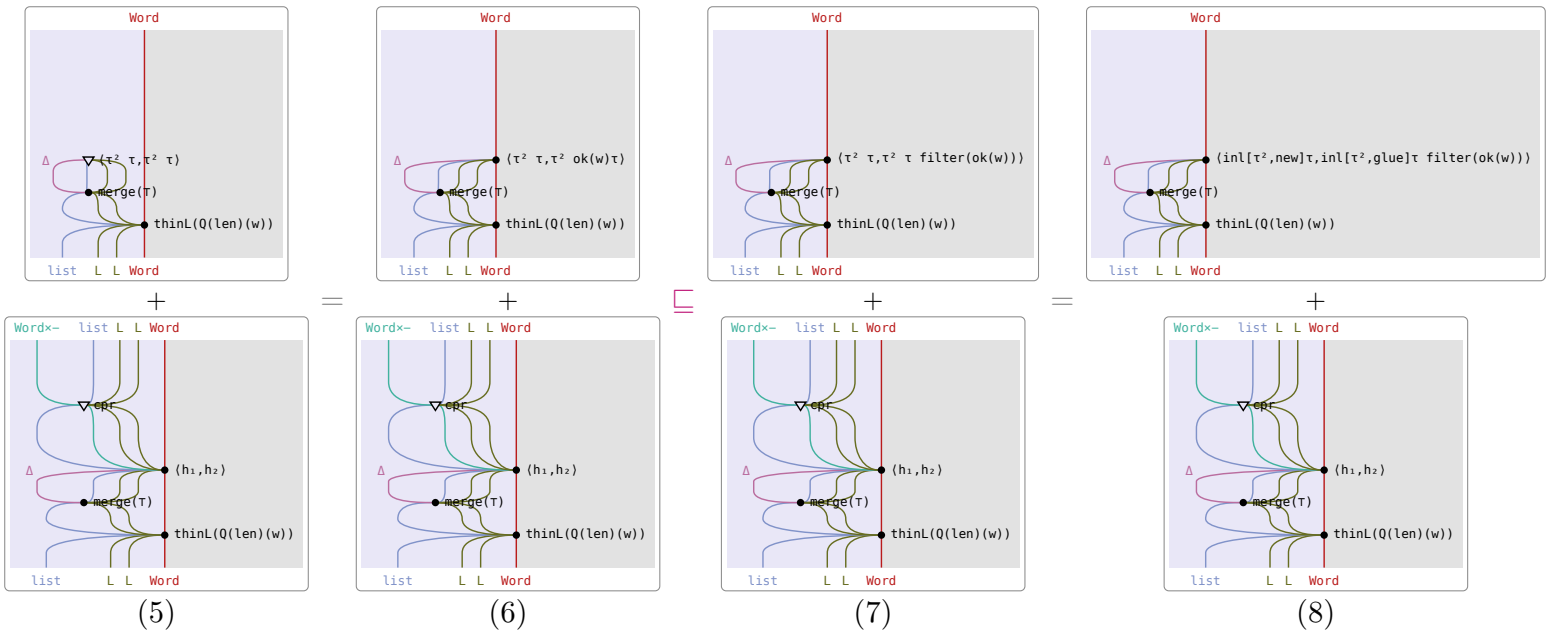
(∀a. len(a) ≤ w) ⇒ [τ² τ,cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))] ⊑ cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)) (8.5.2a)

the fold's body splits into a base case **start** for a single word and a step that extends each kept layout by **new** and by **glue**

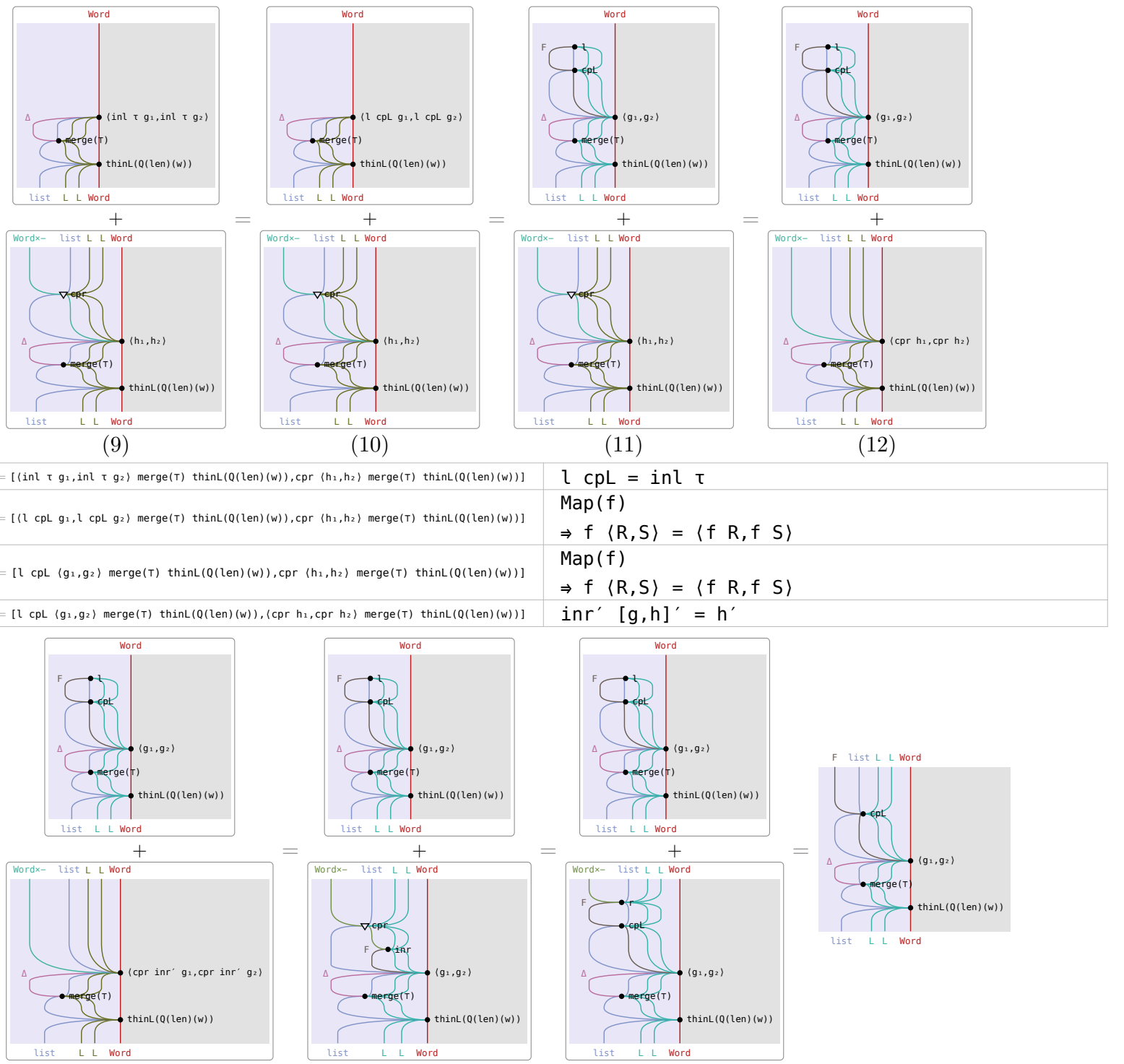


1	[τ² τ,cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	τ² = τ²
2	= [τ² τ,cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	𝟙 ⊑ Q(len)(w)
3	⊑ [τ² ⟨τ,τ⟩ cat thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	merge(T) = cat

4	= [τ² ⟨τ,τ⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩



5	= [⟨τ² τ,τ² τ⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	(∀a. len(a) ≤ w) ⇒ τ² ok(w) = τ²
6	= [⟨τ² τ,τ² ok(w) τ⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	Coreflexive(ok(w))
7	⊑ [⟨τ² τ,τ² τ filter(ok(w))⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	inl [g,h] = g
8	= [⟨inl [τ²,new] τ,inl [τ²,glue] τ filter(ok(w))⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	τ R′ = R τ



9	= [⟨inl τ g₁,inl τ g₂⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	l cpL = inl τ
10	= [⟨l cpL g₁,l cpL g₂⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩
11	= [l cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)),cpr ⟨h₁,h₂⟩ merge(T) thinL(Q(len)(w))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩
12	= [l cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)),⟨cpr h₁,cpr h₂⟩ merge(T) thinL(Q(len)(w))]	inr′ [g,h]′ = h′

(13)	(14)	(15)	(16)
13	= [l cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)),⟨cpr inr′ g₁,cpr inr′ g₂⟩ merge(T) thinL(Q(len)(w))]	Map(f) ⇒ f ⟨R,S⟩ = ⟨f R,f S⟩	
14	= [l cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)),cpr inr′ ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w))]	r cpL = cpr inr′	
15	= [l cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w)),r cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w))]	l T = R ∧ r T = S ⇒ T = [R,S]	
16	= cpL ⟨g₁,g₂⟩ merge(T) thinL(Q(len)(w))		

§8.6. Bitonic tours

(8.6a)

	definition	type	meaning
1	FX = L + E × X	𝒜→𝒜	here L ≜ City×City and E ≜ City: the last two cities, or a city and what is built from the rest
2	Journey(City) ≜ μF	Type	a list of at least two cities, the last two held as a pair
3	Tour(City) ≜ Journey(City) × Journey(City)	Type	the outward journey and the return journey
4	head[a, b] = a, head([a]⨾a') = a	Journey(City)→ City	the first city of the journey
5	next[a, b] = b, next([a]⨾a') = head(a')	Journey(City)→ City	the second city of the journey
6	tail([a, b]) = [b], tail([a]⨾a') = a'	Journey(City)→ L(City)	the journey without its first city, a non-empty list
7	[a]⨾[c] = [a, c], [a]⨾[b]⨾xs = [a]⨾[b]⨾xs	L(City)→ Journey(City)	a put in front of a non-empty list, a journey
8	replaceHead = (𝟙 × tail) ⨾	Journey(City)→ Journey(City)	replace the first city of the journey by a
9	outcost[a, b] = tc(a,b), outcost([a]⨾a') = tc(a,head(a')) + outcost(a')	Journey(City)→ Int	the cost of travelling the journey forwards
10	incost[a, b] = tc(b,a), incost([a]⨾a') = tc(head(a'),a) + incost(a')	Journey(City)→ Int	the cost of travelling the journey backwards
11	cost(t) ≜ outcost(π₁(t)) + incost(π₂(t))	Int	the outward journey forwards plus the return journey backwards
12	start(a,b) = ([a, b],[a, b])	City×City→ Tour(City)	the tour of the last two cities, both journeys being that one edge
13	dropl(a,t) ≜ (replaceHead(a,π₁(t)),[a]⨾π₂(t))	Tour(City)	a replaces the first city of the outward journey and is put in front of the return
14	dropr(a,t) ≜ ([a]⨾π₁(t),replaceHead(a,π₂(t)))	Tour(City)	a is put in front of the outward journey and replaces the first city of the return
15	[start,dropl](l(p)) = start(p), [start,dropl](r(q)) = dropl(π₁(q),π₂(q))	F(Tour(City))→ Tour(City)	start at the last two cities, then add each city by dropl
16	[start,dropr](l(p)) = start(p), [start,dropr](r(q)) = dropr(π₁(q),π₂(q))	F(Tour(City))→ Tour(City)	start at the last two cities, then add each city by dropr
17	tour ≜ ⦇[start,dropl] ∪ [start,dropr]⦈	μF→Tour(City)	every bitonic tour of the cities
18	R ≜ cost ≤ cost°	Tour(City)→ Tour(City)	t costs no more than t′
19	next₂(t) ≜ (next(π₁(t)),next(π₂(t)))	City×City	the second cities of both journeys
20	head₂(t) ≜ (head(π₁(t)),head(π₂(t)))	City×City	the first cities of both journeys

definition	type	meaning
`Q ≜ R ∩ (next₂ next₂°) ∩ (head₂ head₂°)`	`Tour(City)→ Tour(City)`	t costs no more than t′, and both journeys of t and t′ share their first two cities

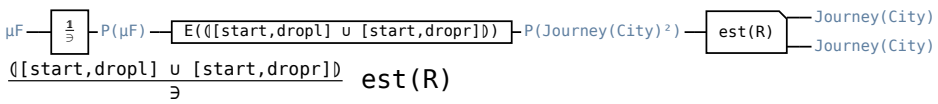
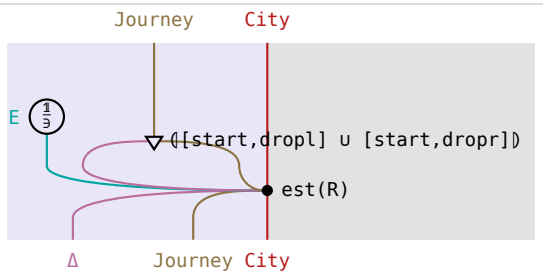
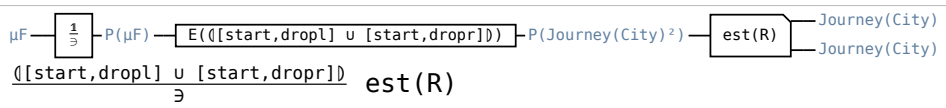
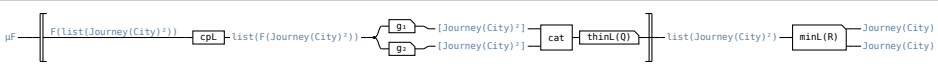
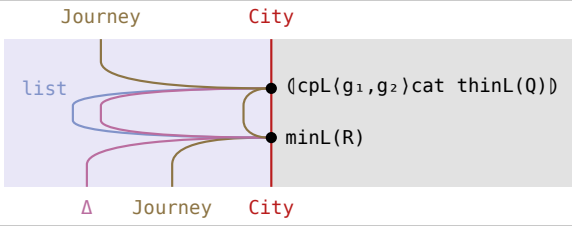
(8.6b)

	the law	what it says
1	`(𝟙×R) droplᵉdropl R` FALSE `(𝟙×R) droprᵉdropr R` FALSE	neither drop is monotonic on `R`: the two edges it adds and removes depend on `head` and `next` of both lists
2	`F(Q) [start,dropl] ⊑ [start,dropl] Q` `F(Q) [start,dropr] ⊑ [start,dropr] Q` ,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
3	`F(T) [start,dropl] ⊑ [start,dropl] T` `F(T) [start,dropr] ⊑ [start,dropr] T`	both algebras are monotonic on `T`, so `T` is the sort order `P`

§8.6.1. Derivation

`(|cpL ⟨g₁,g₂⟩ cat thinL(Q)|) minL(R) ⊑ (|[start,dropl] ∪ [start,dropr]|)/∋ est(R)` (8.6.1a)

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit	Hinze–Marsden
 `(|[start,dropl] ∪ [start,dropr]|)/∋ est(R)`	
=  `(|[start,dropl] ∪ [start,dropr]|)/∋ est(R)` `tour ≜ (|[start,dropl] ∪ [start,dropr]|)`	
⊒  `(|cpL ⟨g₁,g₂⟩ cat thinL(Q)|) minL(R)` binary thinning theorem, at `P≜T` with `merge T=cat`, `Q` from `F(Q) [start,dropl] ⊑ [start,dropl] Q` , `F(Q) [start,dropr] ⊑ [start,dropr] Q`	
= `(|[start τ,cpr ⟨dropl′,dropr′⟩ cat thinL(Q)]|) minL(R)` `cpL=τ+cpr`, `gᵢ=[start′,dropᵢ′]` — `start(a,b) = ([a, b],[a, b])` , `dropl(a,t) ≜ (replaceHead(a,π₁(t)),[a]⧺π₂(t))` ; quadratic, two tours added per step	